ACS APPLIED ELECTRONIC MATERIALS

pubs.acs.org/acsaelm Review

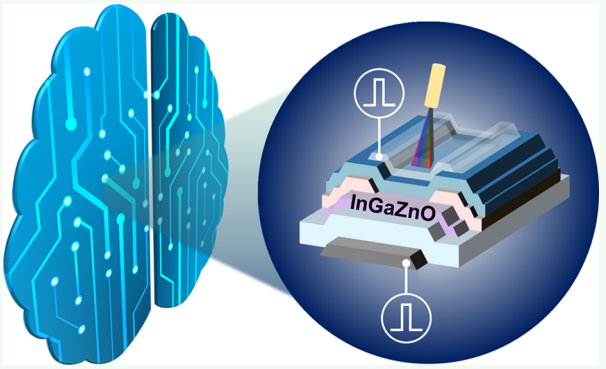
Amorphous InGaZnO (a-IGZO) Synaptic Transistor for Neuromorphic Computing

Yuseong Jang, Junhyeong Park, Jimin Kang, and Soo-Yeon Lee*

Cite This: ACS Appl. Electron. Mater. 2022, 4, 1427−1448 Read Online

ACCESS | Metrics & More | Article Recommendations

ABSTRACT: Brain-inspired neuromorphic computing emulates the biological functions of the human brain to achieve highly intensive data processing with low power consumption. In particular, spiking neural networks (SNNs) that consist of artificial synapses can process spatiotemporal information while enabling energy-efficient neuromorphic computations. Artificial synapses are a key element of sophisticated neuromorphic hardware, so a significant amount of research has been conducted to develop various materials and device structures. Of these, we assess amorphous InGaZnO (IGZO)-based synaptic transistors that have exhibited properties suitable for emerging hybrid optoelectronic neuromorphic systems. Here, we describe the fundamental principles of neuromorphic computations, neuron circuits, and synaptic devices according to recent studies. IGZO-based transistors are discussed, from their material properties to various device physics for electronic- and/or photonic-neuromorphic systems with extraordinary biological emulations.



KEYWORDS: *brain-inspired, neuromorphic computing, synaptic transistor, amorphous oxide semiconductor, InGaZnO (IGZO), hybrid optoelectronic synapse*

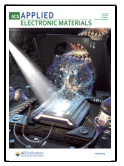
Downloaded via KYUNGPOOK NATL UNIV on November 4, 2025 at 06:05:52 (UTC).
See https://pubs.acs.org/sharingguidelines for options on how to legitimately share published articles.

1. INTRODUCTION

Artificial intelligence (AI), 5G/6G, Internet of Things (IoT), augmented and virtual reality (AR/VR) technologies, robotics, and autonomous vehicles are causing a seismic shift in our daily lives and are producing new user experiences. These technologies undertake massive data processing that requires a high operating power. However, in conventional computing systems, memory and central processing units (CPU) are separate, so power consumption and processing are inefficient as data are being transferred back and forth between the memory and CPU, namely, the von Neumann bottleneck. As an alternative, recently emerging "neuromorphic" systems, which conduct "in-memory" computing, simultaneously perform computations and data processing.[1] Neuromorphic computing mirrors a human brain, which in turn consists of a neural network of $\sim 10^{11}$ neurons and $\sim 10^{15}$ synapses,[2] and consumes only 20 W to process and transmit information.[3] Many studies have reported on software development, but these still suffer from a high dependency on von Neumann architectures that are power-inefficient.[4] Considering that AlphaGo employed 1202 CPUs and 176 graphic processing units (GPUs) and consumed approximately 170 kW for relatively simple calculations,[5] it is reasonable to consider that developing brain-inspired hardware is a task beyond state-of-art computing systems. In the human brain, a postsynaptic neuron receives spikes from presynaptic neurons through synapses with a given weight, and it consequently performs an analogue computation. Neuromorphic systems reproduce this function using a crossbar array with nonvolatile multilevel memristors.[6−12] Various biomimicry functions have also been reported, including recognition, emotion, and recall. Although two-terminal memristors can offer a simple and scalable structure, synaptic weight modulation and computing cannot be simultaneously conducted.[13] On the other hand, a transistor with three or more terminals offers a much simpler form of weight control and computing because programming and read can be separated.[14] Recent research on transistor-based neuromorphic hardware implementations has primarily implemented electronic stimulation to emulate biological functions because adjacent, neighboring neurons in the human brain communicate via voltage spike inputs and outputs. However, a human can also recognize objects through the retina via optical inputs, so a neuromorphic system

Special Issue: Early Career Forum

Received: November 3, 2021
Accepted: January 19, 2022
Published: February 1, 2022



© 2022 American Chemical Society

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

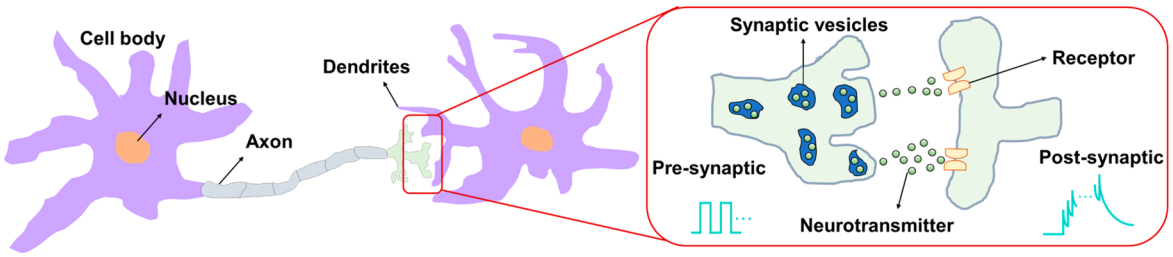


Figure 1. Schematic illustration of the human biological system.

employing photonic stimulation has recently drawn much more attention.[15−20] Photonic stimulation can provide a much lower power consumption and faster processing speed than the electrical stimulation. For example, researchers at IBM recently presented photonic in-memory computing devices using a Si_3N_4 waveguide and phase changing material.[17,18] In this regard, optoelectronic hybrid neuromorphic devices that combine electronic and photonic stimulations together are remarkably rising and demonstrated by many materials, such as silicon,[21] two-dimensional (2D) transition metals,[22] halide perovskite,[23] organics,[24] and amorphous oxide semiconductor (AOS).[25−27] Silicon provides reliable performance with mature complementary metal oxide semiconductor (CMOS) process technologies. However, CMOS devices possess limited capabilities to emulate the analogue functions of the human nervous system since it has a high switching performance that is more suitable for digital computing.[28] Halide perovskite has emerged as a candidate material for use in various applications (solar cell, LEDs, and memory) due to its defect-tolerance, bandgap tunability, and fast ion migration.[29−31] However, high-quality halide perovskite is generally synthesized based on the solution process which results in inferior reproducibility and stability against ambient conditions and also makes use of toxic materials, mainly lead (Pb), all of which make it difficult to commercialize. Among AOS, IGZO has already been commercialized as an n-type channel for an analogue-based driving thin film transistor (TFT) to finely control the current in organic light-emitting diodes (OLEDs) to achieve 256 grayscale.[32] Moreover, IGZO exhibits high electron mobility, even in the amorphous structure, as well as a much lower leakage current than silicon due to a higher band gap (~3.4 eV).[33] In other words, IGZO shows a sensitive spectral response under an ultraviolet (UV) wavelength, resulting in a persistent-photoconductivity (PPC) effect that is detrimental to drive TFT.[34] Therefore, an additional shield structure is inserted to inhibit light irradiation into IGZO TFT.[35] Conversely, this PPC effect, the most particular characteristic of IGZO, meaning that the conductance level is maintained for a while, can be advantageous for optoelectronic synaptic devices since it can emulate biological functions, such as long-term potentiation (LTP). Recently developed optoelectronic neuromorphic architectures generally consist of separate photodetector and synaptic devices,[36,37] which is inefficient in terms of power consumption and scalability. Therefore, a single synaptic device employing light-responsive IGZO is expected to successfully provide a highly power-efficient optoelectronic neuromorphic system.

In this paper, we introduce the fundamentals of artificial neurons and synapses. Next, the material properties and photoresponsivity of IGZO, the device physics, and recent research on multiterminal transistors for neuromorphic applications are reviewed. Future perspectives are then discussed in the last section.

2. NEUROMORPHIC HARDWARE IMPLEMENTATION

Artificial synapses and neurons are the essential components to implement neuromorphic hardware. An artificial synapse is a bridge between the artificial neurons that deliver action potentials, and an artificial neuron is responsible for collecting and integrating spikes transmitted from adjacent neurons through the artificial synapses. In this chapter, we briefly introduce the fundamentals of artificial neurons and synapses.

Artificial Neuron. A single neuron is an elementary constituent of the human brain, and it is segmented into three major components: dendrites, a soma (cell body), and an axon (Figure 1).[38] Dendrites, a branched tree-like structure, are responsible for receiving presynaptic inputs transmitted from the adjacent neuron's axon terminals. The axon is connected to the cell body with a thin and long fibrous structure that transmits electrochemical signals passed from the presynaptic dendrites to the next neuron's postsynaptic dendrites. The gap between adjacent neurons is the synapse. The soma carries the cell nucleus and processes information by integrating presynaptic inputs collected at the dendrites, and it finally determines whether the neuron fires a new spike or not.

The cell body is surrounded by a membrane with a resting potential as the difference between the inside and outside of the cell. However, the membrane potential generally changes by inputs integrated from the axons, which are excitatory and inhibitory inputs, and these then induce an excitatory postsynaptic current (EPSC) or inhibitory postsynaptic current (IPSC). When the total quantity of presynaptic inputs exceeds the membrane potential threshold, the cell body fires a new action potential (or spike). This entire process is known as the integrate-and-fire (IF) mechanism.[39]

An artificial neuron must include capacitors to imitate the role of the membrane of a human nervous system. The most common model is the IF neuron where capacitors collect the sum of current stems from the entire synapses connected to the neuron.[39] When the voltage potential accumulated at the capacitor exceeds the threshold, a voltage spike is generated, followed by an initialization of its potential to zero. Even though the resting membrane potential in a human nervous system is known to be negative, an initialization to zero potential is preferred in an artificial neuron to keep the circuit simple.

From the first report on an IF neuron model, the leaky-integrate-and-fire (LIF) neuron has now been generally accepted as the fundamental model for artificial neurons because "leaky" reflects the decrease in the membrane potential with time that arises from ion diffusion in a biological system.[40] Therefore, the LIF neuron circuit is generally

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

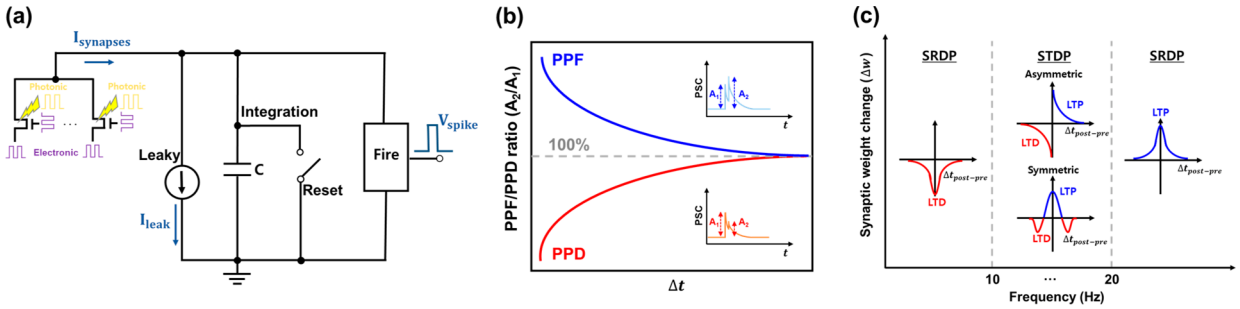


Figure 2. (a) Schematic illustration of an artificial neuron circuit (LIF model) and optoelectronic synapses. (b) Ideal short-term plasticity characteristics (PPF and PPD) as a function of the spike interval (Δt). (c) Frequency-dependent synaptic weight change of STDP and SRDP. STDP dominates at a moderate frequency range (10−20 Hz), whereas the residual region belongs to SRDP. The synaptic weight change reaches its maximum value at a smaller Δt regardless of the frequency.

composed of four parts: leaky, integration, reset, and fire (Figure 2a). Integration is responsible to integrate the incoming current from the synapses. In general, a capacitor is used to store the current in the form of the voltage potential. The capacitance can be adjusted to control the degree to which the potential is stored. Fire generates a spike when the membrane potential reaches a threshold. Reset initializes the membrane potential to zero right after the spike is generated. However, when each part is demonstrated separately, artificial neurons occupy a large area, resulting in a high-power consumption. Several approaches to reduce the power consumption of an LIF neuron have been explored, such as employing a single device operating in a multimode configuration[41] or eliminating the capacitor.[42]

Artificial Synapse. The human brain facilitates learning and information storage through interconnections between the neurons by modulating the "synaptic plasticity",[43] which is the capability of the synapse to alter its strength with respect to the time domain. The synaptic plasticity is generally classified by the characteristic retention time, with long-term plasticity accounting for a synaptic strength maintained for more than some hours, contributing to the learning and memory operations,[44] while short-term plasticity lasts for milliseconds to minutes, being the fundamentals for decoding temporal information.[45] Thus, the difference between long-term potentiation/depression (LTP/LTD) and short-term potentiation/depression (STP/STD) is determined by how long the strengthened/weakened synaptic plasticity is maintained.

Artificial synapses must perform analog computing to imitate the human brain.[46] The synaptic weight can be modulated by a change in conductance of the synaptic device according to (i) external inputs including electronic (polarity, amplitude, and width of voltage spike) and photonic (wavelength, light-intensity, and width of photonic spike) stimulations; and (ii) a temporal correlation between the input and output spikes. Note that the pressure or temperature can also modulate the synaptic plasticity.[47,48] To emulate the spatiotemporal computation of the human biological system, the timing-dependent characteristics of an artificial synapse must be addressed. Paired-pulsed facilitation (PPF) belongs to short-term plasticity, and it describes the PSC amplitude (A_2) triggered by the second stimuli spike enhanced compared to that generated by the first stimuli spike (A_1).[49] The PPF index is defined by $(A_2/A_1) \times 100\%$, and it decreases as a function of the interstimulus interval (Δt) between the spikes (Figure 2b). This PPF index can be fitted by a double exponential decay function.[49]

$$\text{PPF index} = 1 + c_1 \exp(-\Delta t/\tau_1) + c_2 \exp(-\Delta t/\tau_2)$$

where c_1/c_2 and τ_1/τ_2 represent the initial facilitation amplitude and relaxation time of two arbitrary exponential functions. The former is considered as a rapidly decaying function, and the latter is considered as a slowly decaying one so that τ_1 is smaller than τ_2. Paired-pulsed depression (PPD) shows the mechanism to be the same as that of PPF with respect to the time intervals between the spikes, but the only difference lies in that the IPSC amplitude after the second spike descends along the interstimulus interval between the spikes. The PPD index is defined by $(A_2/A_1) \times 100\%$ and can be also fitted by a double exponential decay function.[50]

$$\text{PPD index} = 1 - c_1 \exp(-\Delta t/\tau_1) + c_2 \exp(-\Delta t/\tau_2)$$

The spike-timing-dependent plasticity (STDP) belongs to long-term plasticity and considers the sequence of successive stimuli spikes as well as the time interval.[51] The difference in the arrival time ($\Delta t_{postpre}$) between a pre- and postsynaptic spike within milliseconds determines the relative synaptic weight change (Δw) (Figure 2b). The STDP functions are divided in two types (symmetric or asymmetric) according to the plot of Δw as a function of $\Delta t_{postpre}$. In the case of the asymmetric STDP function,[52] LTP can be demonstrated when a postsynaptic spike arrives after the presynaptic spike within a few milliseconds. If the sequence in the arrival time between the pre- and postsynaptic spike is reversed, the LTD characteristics can be emulated. A short $\Delta t_{postpre}$ induces a significant Δw for both the LTP/LTD characteristics, but a large Δt leads to a small Δw.

The spike-rate-dependent plasticity (SRDP) stands for the long-term plasticity in terms of the frequency (f) of the presynaptic spikes. The LTP can be obtained when a high frequency presynaptic spikes arrive (20−100 Hz), whereas LTD is demonstrated with low frequencies (1−5 Hz) (Figure 2c).[53,54] The synaptic weight change is enhanced at a shorter interstimulus time, $\Delta t_{postpre}$. Although it is still not clear which learning rule dominates at a specific frequency, the general belief is that the STDP learning rule is applied at a moderate frequency (10−20 Hz), while the SRDP learning rule is accepted at the residual region. Precisely, the synaptic plasticity for a specific frequency range does not solely depend on one single learning rule, meaning that both SRDP and STDP contribute to the synaptic weight change.[55]

With respect to physical structure of the artificial synapse, the essential components are a presynaptic receptor and postsynaptic counterpart, so the simplest device architecture

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

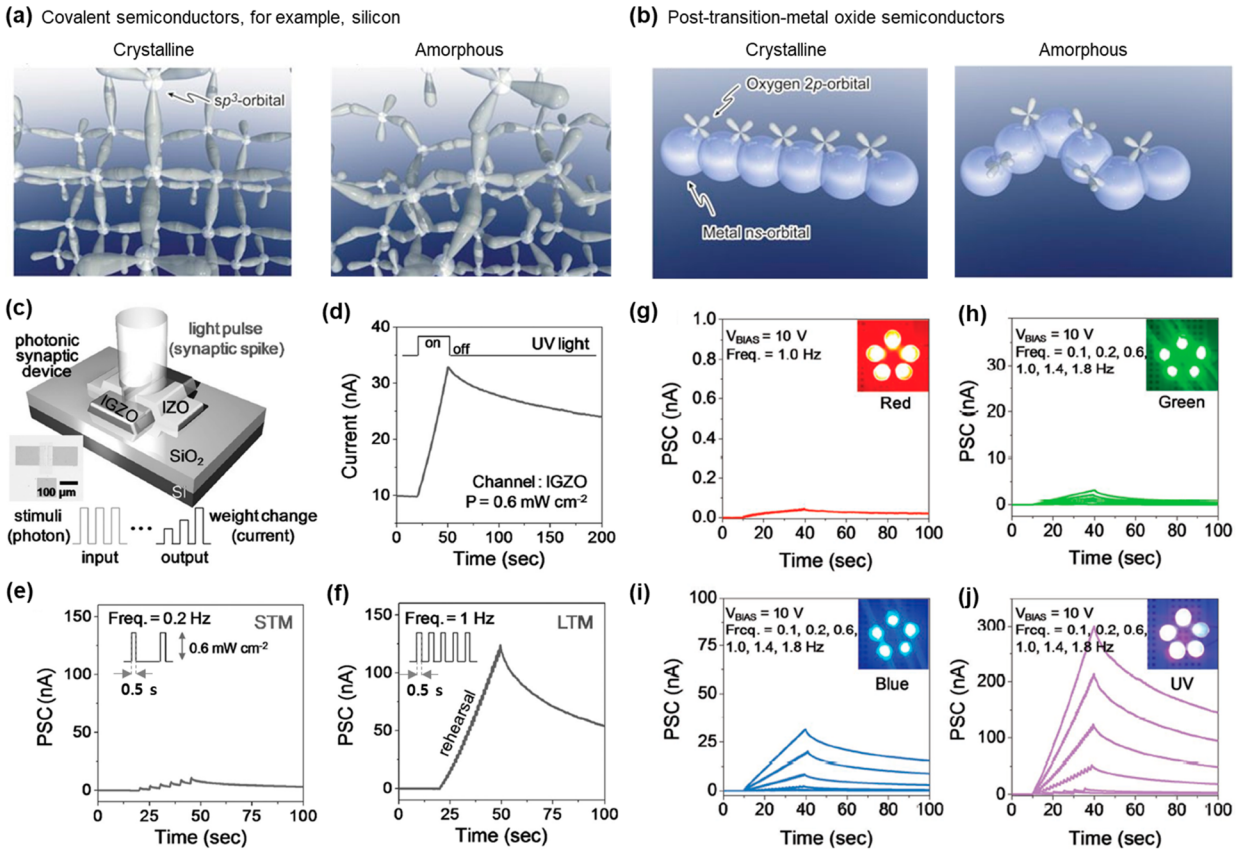


Figure 3. Charge carrier transport mechanisms of (a) covalent semiconductors and (b) post-transition-metal oxide semiconductors in crystalline and amorphous phase. Reprinted with permission from ref 64. Copyright 2004 Springer Nature. (c–j) Photosynaptic plasticity of IGZO. (c) Schematic of the two-terminal synaptic device structure for a photonic neuromorphic system. IZO (InZnO) was used as the electrode to measure the output current. (d) PPC behavior of IGZO against UV light (λ = 380–385 nm, P = 0.6 mW/cm^2, d_t = 25 s). (e) Short-term memory (STM) and (f) long-term memory (LTM) functions with different photonic frequencies of low (0.2 Hz) and high (1 Hz) for UV light. PSC under different frequencies (0.1–1.6 Hz) and wavelengths; (g) red (λ = 620–630 nm), (h) green (λ = 520–530 nm), (i) blue (λ = 460–470 nm), and (j) UV. At a high photon energy and frequency, strong PPC behavior is observed. Reprinted with permission from ref 67. Copyright 2017 Wiley-VCH.

consists of a two-terminal memristor. However, two-terminal memristors cannot simultaneously perform program and read functions, consuming much computation time for online learning.[56] Therefore, a transistor with three or more terminals that parallels program and read can facilitate both online and offline learning, while two-terminal devices are suitable for offline learning.[56]

The features that indicate the performance of an artificial synapse are its linearity, symmetry, conductance gain, or the number of conductance states.[57] Delicate control of synaptic weights should be performed in a highly linear and symmetric way. Conductance gain, also called dynamic range, is defined as the ratio of G_{max} to G_{min}, where G_{max} and G_{min} represent maximum and minimum conductance. Conductance gain can be enhanced with an increase in the number of conductance states, but in the meantime, linearity can be degraded.[58] Regarding fabrication, the artificial synapses should additionally exhibit excellent stability, reproducibility, and uniformity as well as easy processability for a massive array. The influence of these characteristics on the synaptic performance will be discussed with recent studies. Numerous studies have addressed artificial synapses made of various semiconductor materials, and among these, IGZO-based TFT is attractive because it has already been successfully commercialized in large-area displays to supply currents at 256 levels for light-emitting devices with extremely low off-currents. Therefore, we strongly believe that IGZO-based TFT can sufficiently meet the prerequisites of an artificial synapse, including a low power, multilevel operation, and other practical factors for manufacturing.

3. IGZO-BASED ELECTRONIC- AND/OR PHOTONIC-SYNAPTIC DEVICES

We begin this chapter by addressing the material properties and photonic-synaptic plasticity of IGZO, and we then review IGZO-based synaptic transistors that use electronic and/or photonic stimulations with three and four terminals. A four-terminal transistor in which an extra gate node is added, can induce two conduction channels, and therefore can have improvement in electrical properties, including a higher mobility, steeper subthreshold, or enhanced on/off ratio.[59−62] Recent studies on high-performance artificial synapses in conventional TFT structures, flash memory-type configurations, and ferroelectric synaptic transistors are discussed.

Material Properties and Photosynaptic Plasticity of IGZO. IGZO has been mainly applied in industry due to its optical transparency, low processing temperature, and compatibility with various gate insulators.[63−66] Contrary to sp^3 hybridized silicon, the IGZO can withstand mobility degradation under amorphization due to spherically surrounded large s orbitals induced by transition metal cations

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

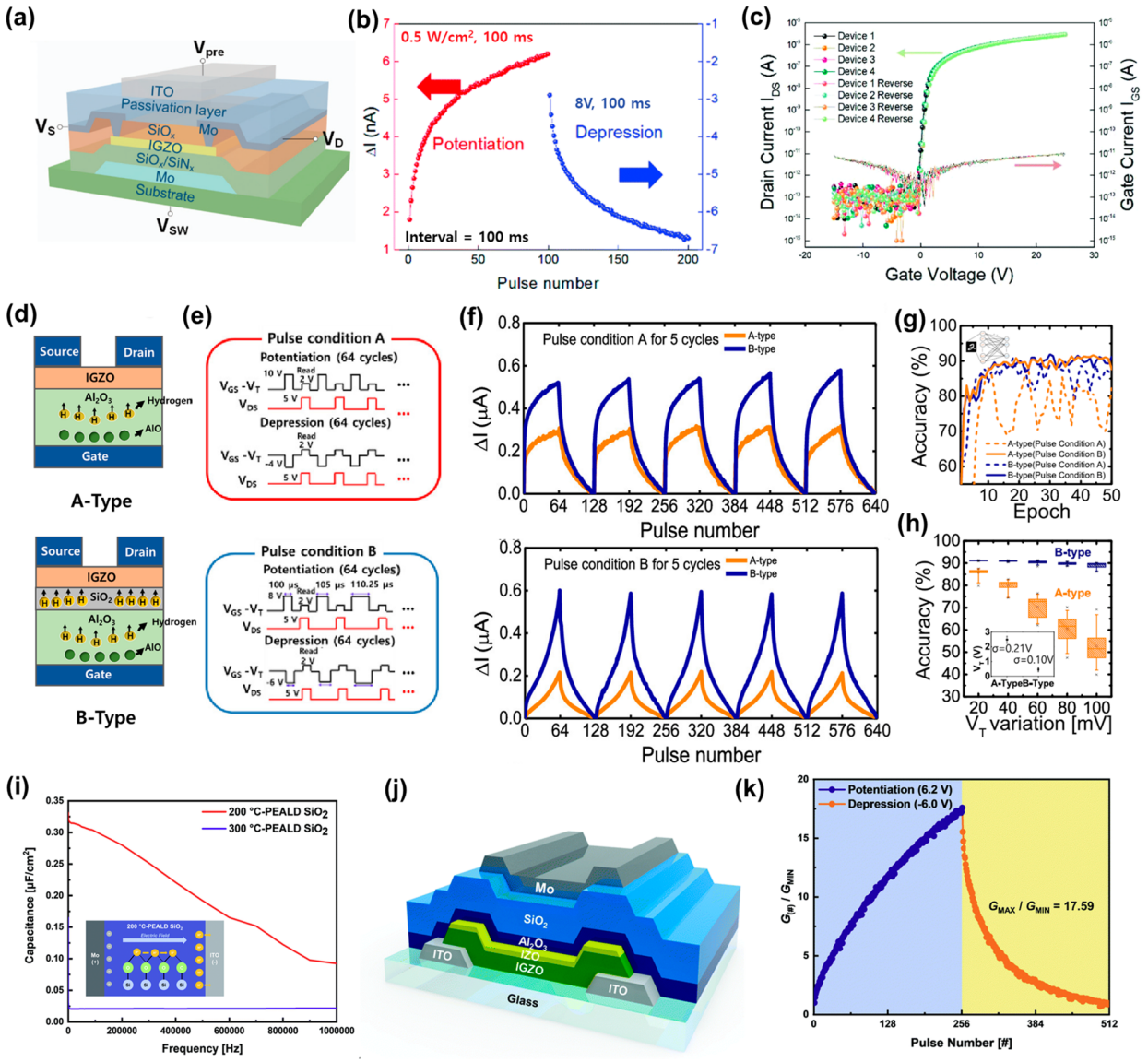


Figure 4. Conventional synaptic transistors. (a–c) Optoelectronic synaptic transistor with four terminals. (a) Device structure of an artificial optoelectronic transistor with four terminals. (b) Potentiation/depression behavior under photonic (P = 0.5 W/cm^2, d_t = 100 ms, Δt = 100 ms)/electronic (V_{pre} = 8 V, d_t = 100 ms, Δt = 100 ms) spikes. V_{pre} (top gate bias) and V_{SW} (1 V) were utilized as presynaptic voltage input and switching voltage input, respectively. (c) Device-to-device variations in transfer curve of synaptic transistors with a V_{SW} sweep measured at V_{DS} = 0.5 V and −0.5 V (reverse). Reprinted with permission from ref 85. Copyright 2019 Royal Society of Chemistry. Electronic synaptic transistor with three terminals using (d–h) IGZO and (i–k) IZO/IGZO coupled with Al_2O_3 and SiO_2. (d) Device structure using a dielectric of Al_2O_3 (upper; A-type) and Al_2O_3/SiO_2 (lower; B-type). (e) Pulse schemes of identical (upper; condition A) and incremental (lower; condition B) pulse width in potentiation and depression. (f) Potentiation/depression behaviors and (g) pattern recognition accuracy according to pulse schemes and devices. (h) Effect of V_{TH} variations on accuracy with different types. Reprinted with permission ref 90. Copyright 2021 American Chemical Society. (i) C–F curves showing frequency-dependent and larger capacitance of 200 °C-SiO_2 (EDL) compared to 300 °C-SiO_2 (conventional). (j) Device structure of a IZO/IGZO synaptic transistor. (k) Potentiation/depression behaviors with each of 256 states and G_{max}/G_{min} = 17.59. The V_{GS} for read was 1.5 V. Reprinted with permission 92. Copyright 2021 Royal Society of Chemistry.

(Figure 3a,b).[64] This feature allows for its carrier transport mechanism independent of crystallinity, exhibiting a high electron mobility of 10 cm^2 V^{-1} s^{-1}.[33] Zn plays a fundamental role as an element that forms the ZnO crystal lattice.

When combined with Ga_2O_3 and In_2O_3, the resulting compound IGZO mostly shows an amorphous phase regardless of relative compositions.[68] It is strongly affected by the differences in the oxygen coordination numbers between the transition metal cations and intrinsic amorphous nature of Ga_2O_3. At the same time, the Hall mobility is assumed to be zero at the maximum compositions of Ga_2O_3[68] and experimentally demonstrated.[69] That is, Ga_2O_3 promotes amorphization and suppresses the carrier concentration of IGZO because the binding energy of Ga_2O_3 is too high to generate a sufficient oxygen vacancy (V_O).[70] In contrast, In_2O_3 has a relatively lower binding energy and can enhance the carrier concentration as well as the electron mobility due to a higher probability of V_O formation.[70] Therefore, InZnO (IZO) without Ga can be used as an electrode.[71]

The key enabler for IGZO to behave as an n-type semiconductor channel in TFT is the intrinsically existing V_O that usually forms during deposition. The oxygen vacancy

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

creates a significant number of localized subgap defect states within the bandgap and, therefore, strongly affects the optoelectronic properties of IGZO.[72] The localized coordination of the oxygen vacancy within the IGZO crystal structure produces shallow donor levels or shallow/deep trap sites.[33] The former accounts for the n-type semiconducting characteristics of IGZO, and the latter is closely responsible for the stability of the TFT against bias, temperature, and illumination stresses. The detailed defect chemistry of IGZO has been well illustrated.[34,73−75] In particular, electrons trapped at deep levels are easily photoionized under illumination with UV light, resulting in PPC before carrier recombination.[76] Although several mechanisms have been proposed to explain the origin of the PPC effect, the general belief is that the neutralization of an ionized oxygen vacancy is hindered because the activation energy was reportedly high.[67] This PPC effect that lasts for more than several hundred seconds becomes dominant under the negative bias in the TFT structure, similar to the assumption that V_O^+ and V_O^{2+} have the lowest formation energy in ZnO under a negative bias.[77] Therefore, the IGZO TFT exhibits a considerable current density at the subthreshold and bias-off region, even under the light-off state, negatively affecting the display operation. Many studies have been reported to suppress the light instability of IGZO, such as modifying the composition,[78] applying a positive gate pulse,[35,79] and inserting an additional shield.[35] Comprehensive strategies to overcome the light instability of IGZO were reviewed by Yoo et al.,[80] and these are consistent with the fact that conventional transistors employ a crystalline semiconductor to enhance device performance and stability. Since the crystalline semiconductor has an extremely low number of subgap states, it is difficult for the channel conductance to be disrupted in response to light.[81,82]

Meanwhile, when a neuromorphic system additionally employs a photonic stimulation, the data transfer speed increases with a low latency coupled with a wide bandwidth. In addition, neuromorphic computing can be implemented in diverse applications by adding photonic stimulation because humans can perceive objects by detecting photons. To attain optoelectronic hybrid neuromorphic systems with suitable characteristics, various materials that possess a moderate number of subgap states have been studied.[27,83,84] From the perspective of synaptic devices, the PPC effect enables IGZO to store the light stimuli for a long time in the form of a conductance. This particular nonvolatile feature gives IGZO great advantages in emulating biological functions, such as LTP, since the specific conductance level can be sustained in the absence of an external optical input.

The first task is to address the synaptic behaviors of IGZO upon photonic stimulation before providing details on the more robust synaptic transistor with three or more terminals. We briefly review the two-terminal synaptic device. Lee et al. presented a pure photosynaptic plasticity of IGZO by subjecting to photonic stimulations with different wavelengths and frequencies.[67] Despite having a conventional device structure for the bottom gate transistor, the current between the IZO S/D electrode was measured without a gate bias (Figure 3c). A strong PPC behavior was obtained when illuminating with a constant UV light (λ = 380−385 nm, P; light intensity = 0.6 mW/cm^2) for 25 s (Figure 3d). Consecutive photonic UV spikes with a shorter duration (d_t = 0.5 s) were applied to emulate the short-term memory (STM) and long-term memory (LTM), respectively, under low (0.2 Hz) and high frequencies (1 Hz) (Figure 3e,f). Moreover, photonic stimulation with different wavelengths (red, green, blue, and UV) and frequencies (0.1−1.8 Hz) obviously governs the photonic-synaptic plasticity (Figure 3g−j). An increase in the photon energy and frequency promotes the PPC behavior, demonstrating LTP behaviors. Therefore, this study showed that synaptic behavior can be effectively enhanced by employing photonic stimulations, not just by electronic ones.

Conventional Synaptic Transistor. In general, nonvolatile multilevel memory function is an essential characteristic for synaptic devices. It can be implemented by using a structure that holds captured charges or a polarization state after removing an electric field. Therefore, employing multiple stacks for charge trapping or ferroelectric materials for remnant polarization is one of the most practical approaches to emulate synaptic plasticity. However, there are some reports that conventional TFTs, believed as independent of nonvolatility, mimic synaptic plasticity. Duan et al. demonstrated various synaptic functions via an artificial optoelectronic synaptic transistor with four terminals.[85] Conventional SiO_x and SiN_x prepared by plasma-enhanced chemical vapor deposition (PECVD) were used for dielectrics. The top gate was for the presynaptic electronic input, and the bottom gate was for applying switching voltage (V_{SW}) to maintain the resting current of the transistor (Figure 4a). The top gate was made of a transparent conductive oxide Sn-doped In_2O_3 (ITO), so the photons were able to reach the channel layer. The potentiation/depression behavior with each of 100 conductance states was emulated by applying photonic/electronic stimulations 100 times, respectively (Figure 4b). The EPSC in response to UV light was obtained via band-to-band generation and ionization of a neutral V_O, while IPSC upon the positive voltage spike to the top gate was attributed to charge trapping at the interface between the top insulator and channel. The demonstrated synaptic plasticity was further utilized to execute visual recognition simulations with the Modified National Institute of Standards and Technology database (MNIST) data sets. The recognition accuracy approaches 95.99% compared to that of 98.69% from the ideal case, which was due to an insufficient linearity and G_{max}/G_{min} ratio in the potentiation/depression behavior. Therefore, further studies should be conducted to enhance the linearity and G_{max}/G_{min} ratio regarding the device process or the structure and driving method. Besides the synaptic performance, highly uniform (Figure 4c) and stable electrical properties were verified, and these can be attributed to the passivation layer that consists of SiO_x and SiN_x. Although device-to-device and cycle-to-cycle variations severely affect the learning accuracy,[46,86−89] many studies seldom report the uniformity and stability of synaptic devices. As such, this study is noteworthy in that uniformity and reliability were considered for synapse implementation. It should be noted that IPSC was induced by charge trapping at the interface between dielectric (SiO_x/SiN_x) and IGZO.

However, the capability to modulate the channel conductance can be reinforced when the mobile ions are incorporated in the conventional dielectrics. Park et al. demonstrated high-performance artificial electronic synaptic transistor with three terminals by using Al_2O_3 and SiO_2, and a modifying pulse scheme.[90] Two types of IGZO TFT utilizing Al_2O_3 (A-type) and Al_2O_3/SiO_2 (B-type) were prepared via low-temperature (80 °C) atomic layer deposition (ALD) of Al_2O_3 and electron-beam evaporation of SiO_2 (Figure 4d).

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

ALD-Al_2O_3 was reported to contain significant amounts of hydrogen when deposited at a low temperature[91] because the thermal energy is insufficient to completely eliminate the remaining hydroxyl groups. Under the positive bias to the gate electrode, these hydrogens move toward IGZO and become ionized. Then, these H^+ ions can act as electron donors, showing a higher channel conductance that results in a negatively shifted threshold voltage (V_{TH}). However, the B-type device exhibited a much larger negatively shifted V_{TH} under a positive bias stress (PBS) compared to that of the A-type device because of a higher concentration of hydrogen in the B-type, as shown by the X-ray photoelectron spectroscopy (XPS) results. Even though a single SiO_2 involved a few portions of −OH bonds (33%), the Al_2O_3/SiO_2 bilayer contained a much higher intensity of hydroxyl groups (61%), indicating hydrogen migration from Al_2O_3. Therefore, a synaptic transistor with the Al_2O_3/SiO_2 bilayer could effectively modulate the channel conductance and induce a large synaptic weight change with the aid of hydrogen. To demonstrate the potentiation/depression behavior with each of 64 states, two pulse schemes of identical (condition A) and incremental (condition B) pulse widths were applied in the potentiation step (Figure 4e). The B-type device exhibited large synaptic weight changes regardless of the pulse conditions, especially due to the hydrogen migration (Figure 4f). Also, linearity improved significantly under pulse condition B independent of the device type. The results obtained using the MNIST data sets to simulate pattern recognition accuracy were also related to the pulse conditions and device type, and the fluctuation that was attributed to severe nonlinearity under pulse condition A could be relieved when using pulse condition B (Figure 4g). A much higher accuracy with less fluctuation was obtained with the B-type device. Furthermore, the effect of V_{TH} variation in B-type devices had a negligible effect on the accuracy (Figure 4h). This interesting study also presented the mechanical flexibility and low energy consumption per spike of 0.18 pJ during the learning steps. From the perspective of device physics, the improved synaptic characteristics in this study are attributed to H^+ ions.

In fact, the movement of these mobile ions can modulate synaptic plasticity since the biological system communicates via neurotransmitters such as K^+, Na^+, and Cl^- ions. A material that involves mobile ions is called an electrolyte. The mobile ions within the electrolyte are drifted depending on the polarity and magnitude of the gate bias, resulting in electric-double-layer (EDL) formation or an electrochemical doping effect.[92,93] EDL refers to the formation of two thin layers with opposite charges at the electrolyte/semiconductor and gate electrode/electrolyte interfaces.[94−96] IGZO TFTs employing EDL have shown very promising device performances of ultralow operating voltage,[66] flexibility,[97] and air stability.[98] The synaptic transistor can emulate the STP characteristic by redistributing the polarized mobile ions in electrolyte under a low gate bias. However, the mobile ions can penetrate the channel under the high gate bias, inducing LTP characteristics. Tremendous efforts have been made to develop the EDL-based IGZO TFTs for electronic- and/or photonic-synaptic devices using polymer,[99,100] ion gel,[101] and solid-state electrolyte.[102−104] Artificial optoelectronic synapses using chitosan and sodium-incorporated Al_2O_3 are introduced in the Advanced Application section.[105−107] As for a solid-state electrolyte, 0.269 fJ per spike, which is lower than that in the human brain (∼10 fJ), was demonstrated by an IZO/IGZO artificial electronic synaptic transistor with three terminals employing SiO_2.[92] A 30 nm-thick SiO_2, prepared by the low-temperature plasma-enhanced atomic layer deposition (PEALD) at 200 °C, was used as an EDL gate dielectric. As discussed in the synaptic transistor using low-temperature ALD-Al_2O_3 (Figure 4d−h), the low-temperature deposition should not complete the chemical reaction, leaving behind the hydroxyl groups. In this paper, the secondary ion mass spectrometry (SIMS) and Fourier-transform infrared (FT-IR) spectroscopy results support that the concentration of hydrogen atoms and hydroxyl groups in SiO_2 increases when process temperature decreases from 300 to 200 °C. The existence of mobile H^+ ions also can be confirmed by measuring the capacitance of the MIM (Mo/SiO_2/ITO) capacitor (inset of Figure 4i). When the electric field is applied, the weakly bonded hydrogen atoms become H^+ ions and accumulate at the SiO_2/ITO interface by jumping from the adjacent oxygen atoms to another. Then, a large capacitance is obtained because the polarization of mobile H^+ ions is coupled with that of electrons. Besides, when SiO_2 contains the mobile H^+ ions, the capacitance strongly depends on the frequency due to ions' lower mobility than electrons' one.[95] The capacitance−frequency (C−F) measurements reveal that 200 °C-SiO_2 has a large and frequency-dependent capacitance, indicating the EDL formation, while 300 °C-SiO_2 shows conventional SiO_2 characteristics (Figure 4i). After IZO-only, IGZO-only, and IZO/IGZO channel TFTs were compared, the IZO/IGZO channel was used to take advantage of IZO's high mobility and IGZO's carrier suppression of IZO. A 10-nm-thick PEALD-Al_2O_3 was additionally inserted as a hydrogen barrier because the IZO transistor with 200 °C-SiO_2 was too conductive. The device structure of the top gate synaptic transistor is shown in Figure 4j, and device-to-device uniformity for seven devices was verified. The device exhibited an anticlockwise hysteresis under the double sweep transfer curve due to the low mobility of H^+ ions. The H^+ ions near the gate electrode screen the gate electric field under the forward sweep, resulting in a positive shift, whereas more electrons are accumulated by the H^+ ions near the channel under the reverse sweep, causing a negative shift. An IZO/IGZO synaptic transistor exhibited the lowest energy consumption of 0.269 fJ per spike and PPF characteristic under the same magnitude of V_{GS} (3.5 V). The potentiation/depression behavior with a G_{max}/G_{min} ratio of ∼17.6 and each of 256 states was obtained by applying 6.2 V/−6.0 V (Figure 4k) along with excellent cycle-to-cycle variation (3.91%) for six cycles. Considering that the highest energy consumption per spike was 2.41 fJ under a V_{GS} of 10 V, the combination of EDL and semiconductor with high mobility and optimum carrier concentration can implement an exceptionally power-efficient artificial synapse beyond the human brain.

The devices could sufficiently demonstrate synaptic characteristics when the dielectric contains significant amounts of the mobile ions during a low-temperature ALD and PEALD process and exhibits general characteristics of EDL. From the fact that the EDL-gated artificial synapse utilizes the movement of ions, the principle of synaptic plasticity modulation agrees well with that of a genuine biological system. However, when applying consecutive high gate voltage spikes to emulate LTP characteristics, the electrolyte should contain a moderate concentration of hydrogen ions because a too high concentration of H^+ ions makes the device conductive. Interfacial trap density is also important because charge

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

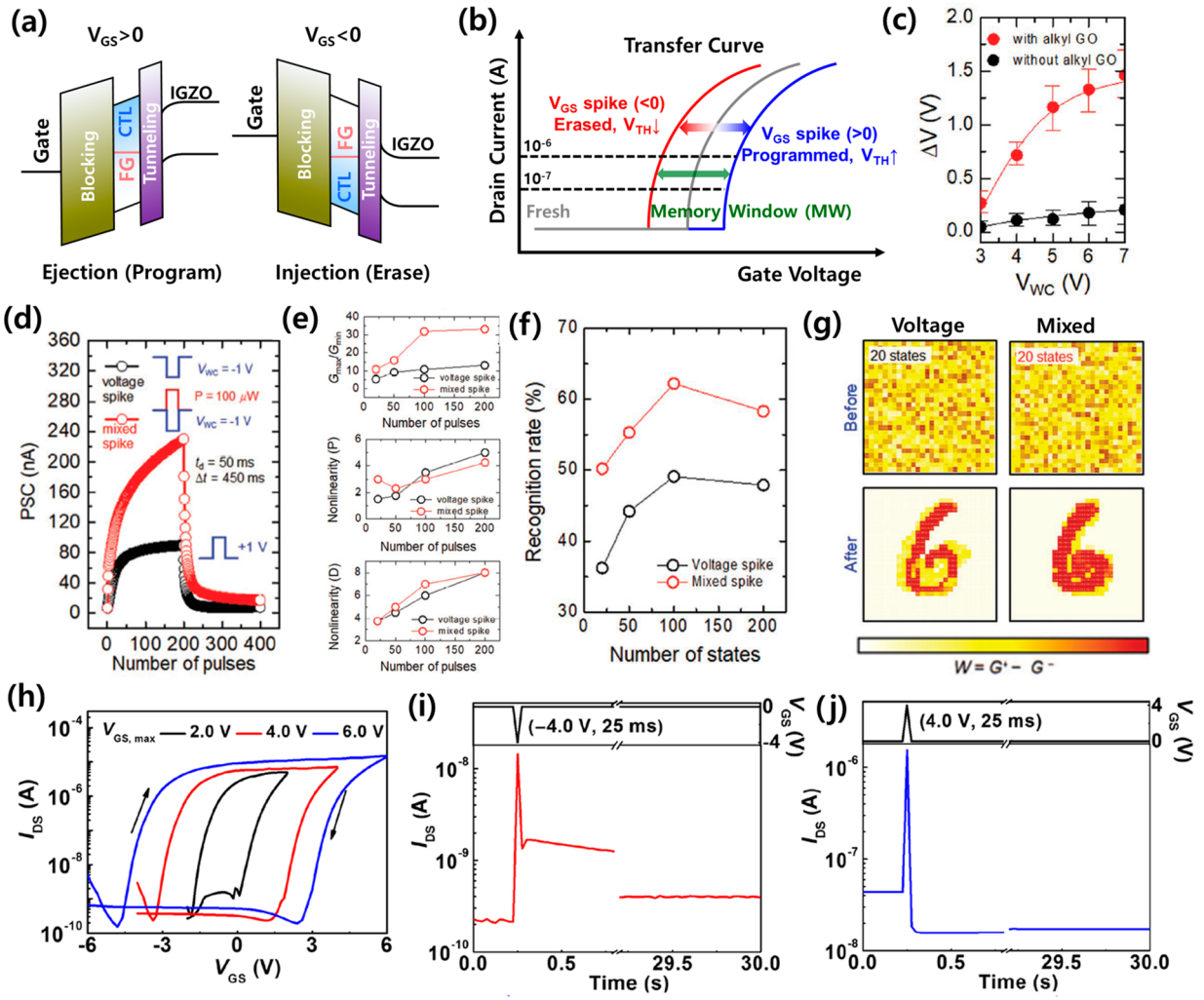


Figure 5. Flash memory-type synaptic transistors. (a, b) Fundamentals of a flash memory device. (a) Energy band diagram under a positive V_{GS} (left) and negative V_{GS} (right), and (b) typical transfer curve. (c–g) Optoelectronic FG-type synaptic transistor. (c) ΔV as a function of gate voltage (V_{WC}; weight control) sweep ranges that can represent the memory window. ΔV was defined at V_{DS} = 1 V and I_{DS} = 10^{-7} A. (d) Potentiation and depression behaviors demonstrated by voltage spikes and mixed spikes (voltage and light). Photonic spikes (λ = 405 nm, P = 7.23 mW/cm^2) were only applied in the potentiation phase. (e) Synaptic parameters of G_{max}/G_{min} (first row), nonlinearity of potentiation (second row) and that of depression (third row) as a function of the number of states. (f) Recognition rate under voltage spikes and mixed spikes as a function of the number of states. (g) Mapping images for the recognition of the letter "6" obtained by voltage and mixed stimulations with each of 20 states. Reprinted with permission ref 58. Copyright 2018 Wiley-VCH. (h–j) Electronic FG-type synaptic transistor. (h) Transfer curve hysteresis with V_{GS} sweep ranges. MW was defined at V_{DS} = 2 V and I_{DS} = 10^{-6} A (i) EPSC and (j) IPSC obtained under negative and positive V_{GS}. Reprinted with permission ref 127. Copyright 2020 IOP Publishing Ltd.

trapping/detrapping causes clockwise transfer curve hysteresis, which impedes the effect of EDL. The mobile ions can be neutralized by the interfacial traps, and electrons in the channel are repelled by the trapped electrons.[104,108,109] The concentration of hydrogen atoms and interfacial traps of ALD- and PEALD-Al_2O_3 are affected by the oxidants and plasma time.[110] Additionally, the deposition temperature should be chosen carefully since the surface roughness of thin film that varies with the thermal energy influences the interfacial traps.[111] It was reported that plasma treatment can effectively reduce these interfacial traps.[104] Other vacuum processes such as PECVD and sputtering can also induce EDL modulation within conventional dielectrics of SiO_2,[112,113] Al_2O_3,[114,115] and HfO_2.[104]

The papers introduced in this section show that IGZO could successfully emulate synaptic behaviors when a double gate structure or mobile ion-contained dielectric is employed. Since control of the channel's electric field or dielectric's ion concentration is important, transistors with conventional dielectrics using semiconductors other than IGZO can also implement synaptic characteristics. In fact, an organic semiconductor has realized outstanding work using conventional dielectrics of SiO_2 and ALD-Al_2O_3 without an EDL effect.[116] This synaptic transistor consumed 0.11 fJ per photonic spike (λ = 365 nm, 0.5 μW/cm^2), assisted by a photosensitive halide perovskite. The channel was 2,7-dioctyl[1]benzothieno[3,2-*b*][1]benzothiophene (C8-BTBT) with high crystallinity, known to have an optical bandgap of 3.6 eV,[117] and a mobility of ~10 cm^2 V^{-1} s^{-1}.[118] However, this organic semiconductor was prepared by a solution process. Despite the advantages of the solution process, such as low cost and easy processability, the vacuum process offers better opportunities for reliable and stable devices. IGZO permits the vacuum process and exhibits similar mobility even in the amorphous phase as well as UV-

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

sensitive light absorption. Therefore, we believe that IGZO is suitable for reliable optoelectronic synaptic transistors as the channel.

Flash Memory-type Synaptic Transistor. Artificial synapses should maintain the updated weight for a long time to emulate long-term plasticity and should also communicate via a voltage pulse within ms.[119] Therefore, nonvolatile and fast responses are the most crucial prerequisites. Studies on IGZO TFT for nonvolatile memory application have been reported by using heterojunctions and thermal treatment.[120−122] A flash memory is another interesting approach, which is a mature nonvolatile memory technology that benefits from high integration density and a fast program/erase speed in the ns to ms range.[123] Therefore, flash memory-type transistors can be an excellent candidate for artificial synapse. Flash memory operates by accumulation/depletion of charge carriers at the charge storage layer sandwiched between the blocking and tunneling layer. There are two types of charge storage layers: floating gate (FG) and charge trap layer (CTL). Figure 5a presents the energy band diagram of the flash memory device. The channel conductance is modulated according to the polarity and magnitude of the voltage spikes applied to the gate terminal (V_{GS}). A control gate bias results in band bending, and the electrons (holes) are transported back and forth between the semiconductor channel and charge storage layer across the tunneling layer with a thickness of a few nanometers. Then, the carriers are accumulated at the charge storage layer because the blocking layer that is tens of nanometers thick prohibits the escape of the carriers. IGZO-based TFT exhibits n-type characteristics, with a positive V_{GS} pulse, so the electrons that exist in IGZO are transported to the charge storage layer, and the channel conductance decreases, resulting in a positive V_{TH} shift (programmed) (Figure 5b). In contrast, under a negative V_{GS} pulse, the electrons accumulated at the charge storage layer move toward IGZO, and the channel conductance increases, with a negatively shifted V_{TH} (erased). Depending on the polarity, amplitude, and duration time of the V_{GS} pulse, the difference in V_{TH} (ΔV_{TH}) for the transfer curve produce a memory window (MW) that represents how many carriers can be trapped/detrapped at the charge storage layer. In general, MW is defined as ΔV_{TH} when the drain current is at 0.1−1 μA. The synaptic plasticity can be modulated by changing the channel conductance of flash memory. When a small read voltage (V_{read}) and V_{DS} are applied, the measured current (I_{DS}) represents postsynaptic currents in the form of EPSC or IPSC.

The FG-type IGZO TFT for memory generally utilized metal nanoparticles as a FG.[124,125] The electronic synaptic transistor that uses IGZO and metal nanoparticles was proposed by Kim et al.[126] This study showed high-performance synaptic functions in an MNIST simulation. Unlike conventional FG-type TFT, this structure was impressive in that the metal nanoparticles were inserted between the dielectric and IGZO.

An artificial optoelectronic synaptic transistor with an FG-type structure was demonstrated by Sun et al.[58] Graphene oxide (GO) nanosheets with long alkyl chains were placed between a blocking layer of ion gel, and IGZO to serve as the FG. The charge trapping capability was verified by MW with an increasing V_{GS} range (Figure 5c). The device without GO showed short-term plasticity in both EPSC and IPSC, but long-term plasticity was demonstrated when GO was inserted. Furthermore, photonic stimulations (λ = 405 nm, P = 7.23 mW/cm^2) were simultaneously applied with a negative V_{GS} to enhance the potentiation behavior (Figure 5.d) because charge detrapping from FG is accelerated by photogenerated holes within IGZO in the potentiation phase. Although the depression behavior was emulated only by a positive V_{GS}, optoelectronic (mixed) stimulations exhibited the highest G_{max}/G_{min} of 33 after 200 pulses (Figure 5e). The nonlinearity of the potentiation/depression tended to increase with the pulse numbers. Furthermore, the recognition rate was determined as a function of the number of states (Figure 5f). From Figure 5e,f, a high G_{max}/G_{min} enormously improves the recognition rate, even with a poor linearity. However, when G_{max}/G_{min} has already been saturated after 100 states, a lower recognition rate was ascribed to nonlinearity. The letter "6" was clearly perceived under the optoelectronic simulations compared to electronic stimulations with each of only 20 states (Figure 5g). This result is an impressive visual image that represents the strength of the optoelectronic simulations on the recognition performance. The most noteworthy point is that the optoelectronic simulations highly enhanced the recognition rate due to the dramatic increase in G_{max}/G_{min} compared to the electronic stimulation.

He et al. prepared an FG-type electronic synaptic transistor by stacking Al_2O_3/ITO/Al_2O_3 as blocking layer/FG/tunneling layer, respectively.[127] The charge trapping performance was identified by a clockwise transfer curve hysteresis (Figure 5h) with increasing V_{GS} sweep ranges. MW varied from 1.2 to 7.3 V. EPSC and IPSC were emulated by applying a voltage pulse of ±4 V. EPSC initially exhibited LTP behavior after removing the negative voltage pulse, and then it instantaneously decreased (Figure 5.i), which was assumed to be due to hole leakage from the FG. In general, when a positive voltage pulse is applied, the electrons at the IGZO channel should be depleted, resulting in IPSC. However, the drain current increased instantly during the positive voltage pulse and ended up with IPSC. After that, PSC increased slightly with time (Figure 5.j), which was also assumed to be due to electron leakage from FG. These results project well that FG is inefficient in achieving a good retention property for memory.[128] However, the potentiation/depression behavior showed that the FG-type IGZO synaptic transistor is also a promising candidate for neuromorphic systems. EPSC increased from 0.2 nA to 228 nA after 32 negative voltage pulses and then decayed to 0.3 nA, which is close to its initial value after 16 positive voltage pulses. The resulting power consumption per spike of the transistor was calculated to be from 0.4 pJ to 286.2 pJ, depending on the pulse scheme. The MNIST simulations to classify three data sets (small digits, large digits, and file types) showed each having a high accuracy of 94.8%, 94.1%, and 93.2%, respectively.

However, the charge leakage in Figure 5i,j was observed in the absence of an electric field and is attributed to direct tunneling through the thin tunneling layer. Direct tunneling can be severe in conductive FG-type devices compared to relatively insulating CTL-type devices because more charge carriers in FG can tunnel through and trigger a higher leakage current. Therefore, the FG-type device requires a thick tunneling layer compared to the CTL-type in flash memory technology so that a high integration density is difficult to be realized.[123] The CTL-type IGZO TFT for memory used insulating Si_3N_4, HfO_2, and Al_2O_3 as a CTL.[62,129−131] However, it was difficult to erase electrically since the hole transport from IGZO to CTL was hindered under the negative

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

Table 1. Compilation of Flash Memory-Type Synaptic Transistors

device stacks (blocking/charge storage layer/tunneling)	L/W (μm)	memory window	memory states	synapse states of potentiation/depression	synaptic functions	ref
ion gel/GO/alkyl chain	10/10	0.27−1.46 V (V_{GS} = ±3−7 V) @V_{DS} = 1 V, I_{DS} = 0.1 μA		200 (−1 V and 7.23 mW/cm^2)/200 (+1 V) d_t = 50 ms, Δt = 450 ms @V_{DS} = 0.1 V	IPSC, EPSC, PPF, MNIST	58
Al_2O_3 (30 nm)/ITO (15 nm)/Al_2O_3 (6 nm)	80/1000	1.2−7.3 V (V_{GS} = ±1−6 V) @V_{DS} = 2 V, I_{DS} = 1 μA		32 (−4 V)/16 (+4 V) d_t = 25 ms @V_{DS} = 50 mV	IPSC, EPSC, MNIST	127
Al_2O_3 (35 nm)/ZnO (20 nm)/Al_2O_3 (8 nm)	10/60	3 V (V_{GS} = 13 V; 1 μs /−8 V; 2 μs) @V_{DS} = 0.1 V	4	1 (−3 V)/1 (+3 V) d_t = 20 ms @V_{DS} = 50 mV	IPSC, EPSC	136
Al_2O_3 (40 nm)/ZnO (30 nm)/Al_2O_3 (10 nm) bottom insulator: Al_2O_3 (40 nm)		2.2−18.2 V (V_{TG} = ±15 to ± 20 V) 16.3−17.0 V (V_{BG} = 2 to −2 V) @V_{DS} = 0.1 V		11 (V_{TG} = −15 V to −20 V)/− d_t = 10 ms ∼1 μs @V_{BG} = 0, V_{DS} = 0.1 V	EPSC, PPF, LTP	137

bias, given that IGZO is naturally an n-type semiconductor. Besides, the charges occupy deep-level trap states within CTL, so they cannot participate in the erasing due to the high energy barrier. Therefore, alternatives to improve the erasing efficiency are adopting IGZO or ZnO as a CTL,[132−134] or assisting light.[135]

Liu et al. demonstrated CTL-type (Al_2O_3/ZnO/Al_2O_3) electronic synaptic transistor that is fast, and low-power erasable without light.[136] The memory device showed a MW larger than 3 V and a good endurance properties up to 10^4 cycles. Stable multilevel memory states with a high linearity were obtained by two pulse schemes with different amplitudes for program of 10 V−13 V and erase of −4 V—12 V, with a width of 1 μs-3 μs. This indicates that the appropriate pulse conditions to reach a desirable level should be determined based on the channel conductance of a previous state. The EPSC and IPSC exhibited LTP and STP characteristics, respectively. EPSC remained well because the ionized oxygen vacancies (V_O^+ and/or V_O^{2+}) accumulated at the Al_2O_3 (tunneling layer)/IGZO interface under the negative bias are difficult to neutralize. In contrast, IPSC rapidly decays to its initial current since trapped electrons at the Al_2O_3/IGZO interface are gradually detrapped.

Ryoo et al. proposed another CTL-type electronic synaptic transistor with four terminals to control the channel conductance effectively using Al_2O_3/ZnO/Al_2O_3 stacks.[137] They first confirmed that MW could be tuned by biasing the top gate and bottom gate with different amplitudes and polarities of the voltage. At V_{BG} = 0 V, MW varied from 2.2 to 18.2 V when V_{TG} sweep range changed from ±10 V to ±20 V. At V_{TG}= ± 20 V, the transfer curve shifted positively with negative V_{BG}, while negatively with positive V_{BG}. MW was reduced from 17 to 16.3 V when V_{BG} changed from −2 V to +2 V. Since an additional gate bias was applied to the channel, the electric field distribution changed so that subthreshold swing (SS) and on-current (I_{ON}) were improved. This double gate configuration offered an excellent opportunity to effectively modulate the channel conductance and operation point. This indicates that the external voltage bias besides the fabrication process can control the device performance in the double gate structure. A linear synaptic plasticity was obtained when incremental step pulse potentiation (ISPP) scheme, which is gradually increased with a pulse train (−15 V—20 V), with a pulse width (1 μs−10 ms) was applied to the top gate. Therefore, appropriate pulse amplitude and width should be chosen for the desirable neuromorphic application. PPF characteristic was also successfully emulated by applying constant V_{TG} at −15 V with a pulse interval in -μs ranges.

Despite the success of flash memory technology, IGZO synaptic transistors with both FG- and CTL-type structures have hardly been reported upon. Moreover, abundant synaptic functions have been demonstrated by the FG-type device compared to the CTL type (i.e., MNIST simulation results were obtained using only the FG type), as summarized in Table 1. We believe this is because the FG type can ignore the trap states of the charge storage layer. Various program and erase mechanisms are possible in flash memory-type synaptic transistors because the amplitude of the voltage pulse is different. FN tunneling is a generally accepted mechanism under high electric field in nonvolatile memory, so that the energy barrier for charge injection/ejection must be considered. However, considering that most IGZO synaptic transistors have been operated under low voltage levels, direct tunneling is also a possible mechanism. In Figure 5i,j, it was assumed that the charge leakage was closely related to the direct tunneling. Therefore, the thickness of the tunneling layer should be concerned. Another factor is interface trap state because it is well-known to accelerate charge trapping in transistors. In fact, interface trap states affected the relaxation time of charge carriers.[138] Moreover, the bulk properties of CTL are the major issues. For instance, trap state level or density rule the tunneling mechanism (trap-assisted tunneling).[139] Not only trap characteristics but also the conductivity of CTL must be controlled. In the case of ZnO, which is CTL mostly deposited by ALD for IGZO-based memory and synaptic transistors, a high deposition temperature must be avoided because of its higher conductivity.[140]

To the best of our knowledge, CTL-type IGZO synaptic transistor using photonic stimulation has yet to be investigated. However, the recent study of CTL-type (Al_2O_3/ZrO_2/Al_2O_3) MoS_2 optoelectronic synaptic transistor achieved 18.3 aJ per electrical spike (−4 V, d_t = 100 ns) and emulated learning-forgetting-relearning behavior by photonic spikes (λ = 350 nm).[141] This result reflect that CTL type can modulate

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

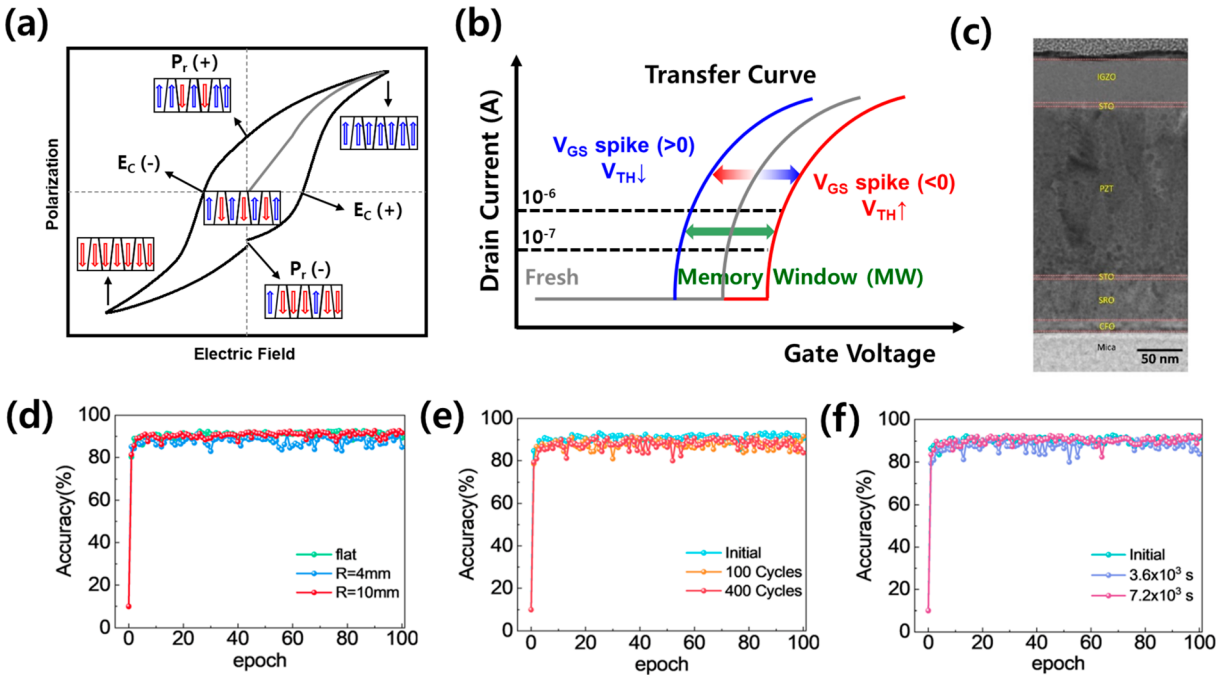


Figure 6. (a, b) Typical ferroelectric characteristics. (a) Polarization−electric field (P−E) hysteresis curve. (b) Transfer curve of FeFET that depends on the polarity of the voltage spike. (c−f) PZT-based ferroelectric electronic synaptic transistor. (c) Cross-sectional transmission electron microscope (TEM) image of an IGZO/PZT-based ferroelectric synaptic transistor. Recognition accuracy according to (d) bending radius, (e) bending cycles, and (f) bending time. Reprinted with permission ref 154. Copyright 2020 AIP Publishing.

photosynaptic plasticity and may even outperform FG type with respect to energy consumption and speed. MoS_2 exhibits semiconducting property with ultrahigh mobility even at a monolayer and its layer-dependent optical bandgap (1.3−1.8 eV) extends to near-infrared (NIR) wavelength.[142] However, 2D materials are commonly prepared by mechanical exfoliation from the crystal, vapor transport-, or CVD-grown substrate, and subsequent wet or dry transfer step.[143] Direct growth by CVD for device fabrication is also possible, but the thickness of 2D materials is hard to control.[144] Therefore, high-quality 2D flakes with favorable thickness and size should be picked out by optical microscope (OM) and atomic force microscopy (AFM) because the optoelectronic properties of 2D materials rely on the number of layer.[145] This takes a long fabrication time compared to IGZO thin film. Furthermore, a usual process temperature in vapor transport and CVD is higher than 500 °C. In contrast, IGZO is generally prepared by sputtering at room temperature with a uniform thickness over a large area. Besides, the multifunctionality of IGZO enables its application to both CTL and channel.[132] Thus, we consider that CTL-type IGZO synaptic transistor is an excellent candidate for optoelectronic synapse with high throughput and integration density.

Ferroelectric Synaptic Transistor. Ferroelectric materials are emerging building blocks that can achieve a low power consumption examined by recent electronics such as ferroelectric tunnel junctions (FeFTJs),[146] ferroelectric field-effect transistors (FeFETs),[147] and negative capacitance (NC) FET.[148] In general, electrical dipoles within the materials are polarized, and the alignment can be reversed according to the direction of the electric field. With an increase in the external electric field across the coercive field (E_c), all domains of ferroelectric materials are completely polarized in the same direction of the electric field. After the electric field is removed, the ferroelectric materials have a remnant polarization (P_r), which is nonvolatile (Figure 6a). Therefore, when implemented in a transistor (FeFET), it is possible to effectively control the channel conductance (i.e., multilevel) with respect to the polarity and the amount of P_r,[149] as illustrated in Figure 6b. In this regard, the ferroelectric transistor can be used to implement power-efficient synaptic devices. Despite efforts to combine various ferroelectric materials and IGZO,[150,151] we are mainly concerned with metal oxides considering the requirements of an artificial synapse.

$Pb(Zr_xTi_{1-x})O_3$ (PZT) in a perovskite crystal structure is a solid solution of two metal oxides, $PbTiO_3$ and $PbZrO_3$. It is known as a popular ferroelectric material because of its high P_r and low E_c.[152] Even though the IGZO/PZT-based FeFET is in an early stage development,[153] Zhong et al. demonstrated its use in a high-performance artificial electronic synapse even at a bending radius of 4 mm.[154] $SrRuO_3$ (SRO), known to mitigate the ferroelectric fatigue of PZT,[155] was used as a gate electrode. $CoFe_2O_4$ (CFO) and two ultrathin $SrTiO_3$ (STO) were embedded for epitaxial growth of SRO and PZT (Figure 6c). The thickness of PZT was larger than 100 nm. The fabricated ferroelectric capacitor (mica/SRO/PZT/Au) exhibited a high remnant polarization of 35 $\mu C/cm^2$. Contrary to the clockwise transfer curve hysteresis in the flash memory-type, anticlockwise hysteresis was observed in FeFET because the remnant polarization of PZT required a much lower gate voltage to reach the same drain current. Therefore, the channel conductance can be modulated by the amount of polarization of the ferroelectric layer. The LTP/LTD behavior (32 level) with a good nonlinearity was obtained under successive voltage spikes of 3.2 V/−3.5 V (d_t = 2 μs, Δt = 200 ms). The MNIST accuracy could be improved by modifying the duration time, interval time, and pulse numbers. The highest recognition accuracy of 94.4% was obtained by the 100 level, close to the

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

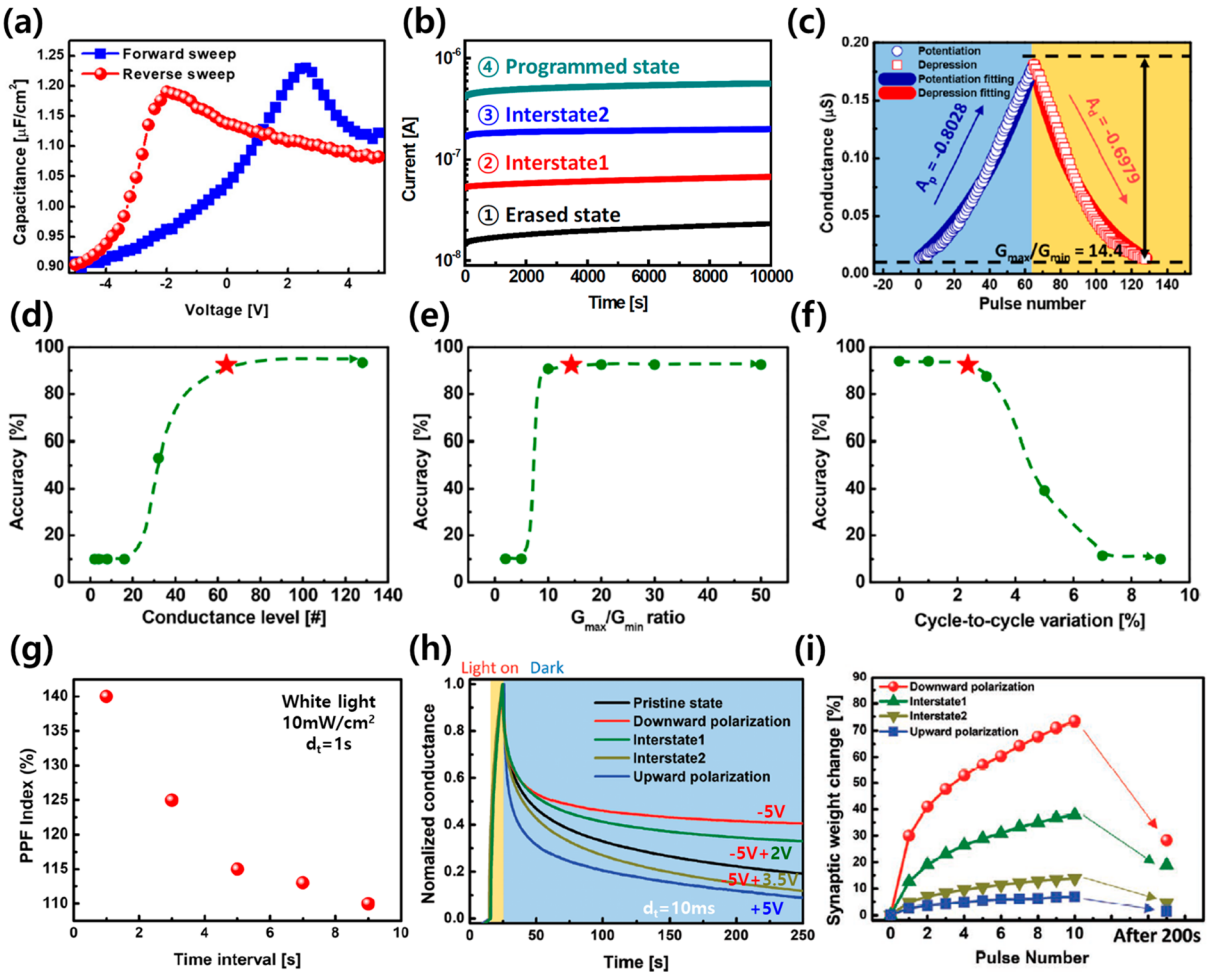


Figure 7. HZO-based ferroelectric synaptic transistor. (a–f) Electronic synaptic transistor. (a) Typical $C-V$ curve that represents the charge trapping capability (b) Retention properties of FeFET under four polarization states. (c) Potentiation/depression characteristics with 64 states, G_{max}/G_{min} = 14.4, and good linearity. Influence of (d) conductance level, (e) G_{max}/G_{min} ratio, and (f) cycle-to-cycle variations on the recognition accuracy for MNIST simulations. Reprinted with permission ref 162. Copyright 2019 American Chemical Society. (g–i) Photonic-synaptic transistor. (g) PPF index of the pristine device before polarization switching under photonic stimulations. Polarization states-dependent (h) conductance relaxation behaviors after photonic stimulation for 10 s, and (i) synaptic weight change decay after 10 photonic pulses (d_t = 2 s, Δt = 1 s), followed by a relaxation time of 200 s. Reprinted with permission from ref 163. Copyright 2020 Wiley-VCH.

ideal of 96%. In addition, an excellent, stable recognition accuracy was obtained under the bending tests with varying conditions for the radius, cycles, and duration times (Figure 6d–f). However, PZT suffers from several drawbacks, such as a too high dielectric constant and thickness, or low integration density.[156]

As an alternative, HfO_2 has attracted significant attention in the semiconductor industry due to its high dielectric constant, ultrathin film thickness, CMOS compatibility, and low-temperature processing, all of which improve scalability.[157] Since the first discovery of ferroelectricity in Si-doped HfO_2 (7–10 nm),[158] undoped- and doped-HfO_2 have been found to exhibit ferroelectricity at a thickness of less than 6 nm[159] and 2 nm,[160] respectively. Several effects have been reported to induce ferroelectricity, including the thermodynamic-kinetic relationship, doping, size, and mechanical stress. For example, when Zr-doped HfO_2 (doping) with a thickness of less than 30 nm (size) surrounded by bottom and top TiN electrodes (mechanical stress) undergoes thermal annealing (thermodynamic-kinetic), ferroelectricity can be obtained.[161]

An artificial electronic synaptic transistor that used IGZO and ferroelectric $HfZrO_x$ (HZO) was proposed by Kim et al.[162] The fabricated MFSM (TiN/HZO/IGZO/Al) capacitor exhibited a P_r of ∼5 $\mu C/cm^2$ and a butterfly loop in capacitance–voltage (C–V) measurement which proves charge accumulation/depletion at the interface between the ferroelectric HZO and IGZO (Figure 7a). An anticlockwise transfer curve hysteresis was observed, and after the initial erasing (−6 V, 30 ms) and incremental programming step over three states, four data levels corresponding to each intermediate polarization state were obtained, and these exhibited superior retention properties of up to 10^4 s (Figure 7b). To emulate the synaptic functions, an increasing bias scheme was applied to the gate electrode. The resulting potentiation/depression characteristics exhibited 64 states with a high G_{max}/G_{min} ratio >10 and good linearity (Figure 7c). In addition, the endurance properties of the conductance levels under potentiation/depression showed extremely low variations in cycle-to-cycle (2.36%) during 100 cycles and device-to-device (3.93%) for 40 devices. These highly reproducible and uniform synaptic characteristics are attributed to the conductance modulation capability that arises from stable and gradual ferroelectric switching under an incremental pulse scheme. The promising characteristics of FeTFT were utilized

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427–1448

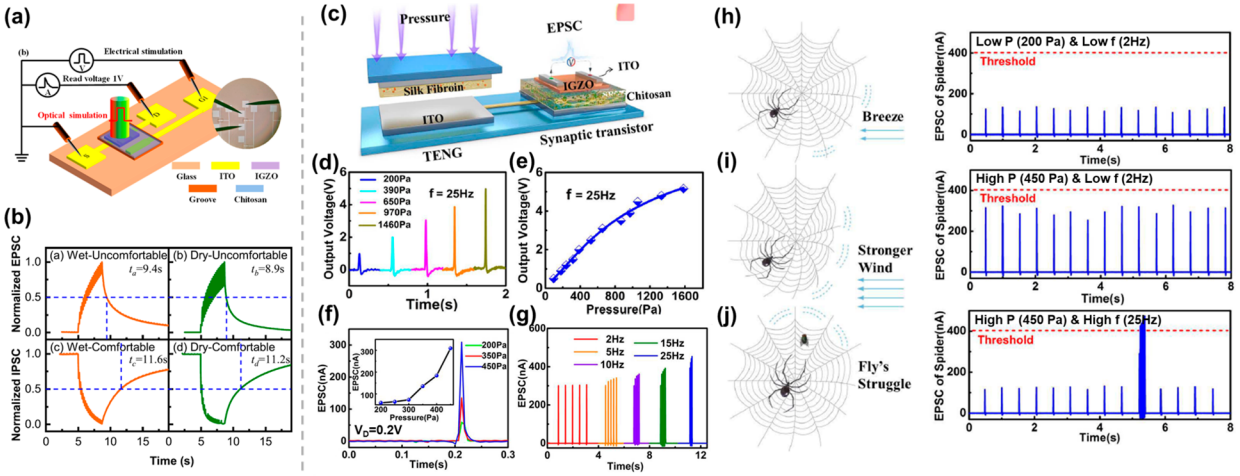


Figure 8. Advanced applications. (a, b) Emotional state simulation. (a) Device configuration of IGZO synaptic transistor and measurement conditions. (b) Normalized synaptic plasticity that decays after positive/negative presynaptic voltage spikes (20 pulses, ±2 V, d_t = 90 ms, Δt = 100 ms) to the gate terminal. "Uncomfortable" and "comfortable" represent the respective emotional states under EPSC and IPSC. Wet and dry refer to the measurement condition that is quiet and noisy. The time constant in each state was determined at the point where the normalized EPSC/IPSC amplitude after the removal of electrical stimulation reaches half of the peak values. Reprinted with permission from ref 105. Copyright 2019 American Chemical Society. (c–j) Tactile perception. (c) Schematic of triboelectric nanogenerator (TENG) and synaptic transistor to demonstrate the tactile sensory behaviors. (d) Output voltage of TENG in the form of a spike. (e) Relationship between the amplitude of the output voltage and pressure at a fixed pressure frequency of 25 Hz. (f) Pressure-dependent and (g) frequency-dependent EPSC behavior. The pressure was applied to TENG and V_D of 0.2 V to the synaptic transistor to read PSC. (h–j) Emulations of the spider's native capacity to identify prey. The threshold for EPSC that can detect the prey was assumed to be 400 nA. Reprinted with permission from ref 107. Copyright 2020 IEEE.

to simulate pattern recognition using the MNIST data sets. The highest accuracy of 91.1% was obtained, and it is comparable to the 94.1% of an ideal synaptic device. The high performance of the ferroelectric synaptic transistor is in agreement with the high accuracy ranges (Figure 7d–f).

The same group proposed an artificial photonic-synaptic transistor with an identical device structure to enhance the PPC characteristics of IGZO through polarization switching of ferroelectric HZO.[163] Figure 7g shows the STP characteristic of the PPF in the absence of ferroelectricity (pristine) emulated by photonic spike pairs (λ = white light, P = 10 mW/cm^2, d_t = 1 s). In addition, the LTP characteristics were also evaluated by repeating photonic pulses with a short time interval. However, the conductance after each photonic pulse did not increase significantly, and it decayed quickly. Therefore, the polarization of HZO was employed to control the conductance of IGZO and thus improve the LTP characteristics. An incremental pulse scheme with four steps (from a negative to a positive bias) induced four intermediate polarization states (from downward to upward). Then, photonic stimulations were applied for 10 s to each of the four states, and the conductance of the synaptic transistor was monitored to identify the relaxation behaviors. A normalized conductance exhibited improved LTP characteristics when the polarization state was closer to a downward direction (Figure 7h). At a downward polarization, the photogenerated electrons are repelled from the interface between IGZO and HZO, so that recombination with ionized V_O is prohibited. In contrast, when upward polarization is induced, the photogenerated electrons are accumulated at the interface, so recombination with ionized V_O is promoted, resulting in a fast conductance decay. Therefore, the suppressed recombination between the electrons and ionized V_O at the downward polarization can enhance the LTP characteristic with a high amplification ratio. As a result, the synaptic weight change after 10 consecutive photonic pulses (d_t = 2 s, Δt = 1 s) followed by a relaxation time of 200 s reached the highest value of 28% at a downward polarization (Figure 7i).

IGZO FeFET for artificial synapses was prepared with HfO_2-based materials less than 30 nm thick, indicating the potential for high integration density. Recently, Cheema et al. successfully demonstrated ultrathin (<2 nm) ferroelectric HZO thin film on Si.[160] This study suggests excellent scalability of the HfO_2-based ferroelectric thin films. Accordingly, many studies on synaptic transistor using Si- and HfO_2-based ferroelectric thin films (<10 nm) were realized.[164,165] For example, a FinFET structure[166] and a low energy of 2 fJ per electrical spike (−0.2 V, d_t = 1 μs)[167] were demonstrated by a Si synaptic transistor using HZO. However, in all studies,[160,164−167] native SiO_2 was formed at the interface between Si- and HfO_2-based ferroelectric thin films. Interfacial SiO_2 formation should be avoided because it is known to cause a severe voltage drop or ferroelectric fatigue.[168] IGZO as an oxide semiconductor can be a suitable solution because it has a clear interface without interdiffusion between the IGZO and HZO.[168−170] The ferroelectricity of HfO_2-based materials is usually obtained via TiN capping layer deposition on HZO and subsequent annealing. Various capping metals (W, Pt, Au, Pd, and Ta) have also been explored.[171,172] Besides, an oxide electrode RuO_2 can solve common side effects (fatigue, wake-up effect, domain pinning, and imprint) of ferroelectrics because it may serve as an oxygen vacancy reservoir, while the reactive TiN can be easily oxidized into TiO_xN_y or TiO_2 at the interface between HfO_2 and TiN.[173] However, the electrode/ferroelectric/electrode structure is a two-terminal device with limited applicability. Synaptic transistors that consist of HZO and IGZO were fabricated by etching the TiN capping layer after inducing ferroelectricity and before IGZO deposition.[162,163,174] This additional etching process is unnecessary because directly deposited IGZO on HZO as a capping

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

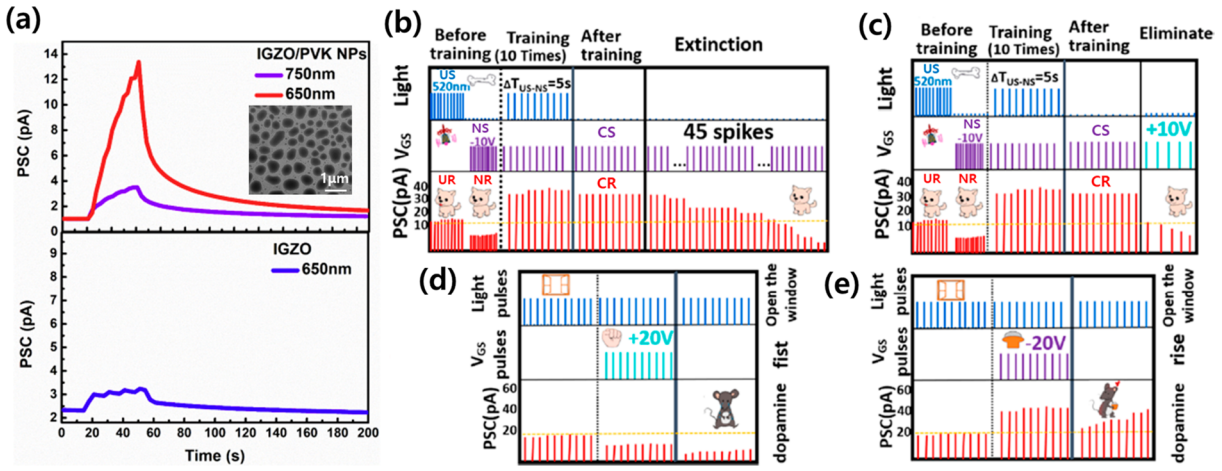


Figure 9. Associative learning. (a) Photosynaptic plasticity under a long wavelength of optoelectronic synaptic transistor with or without PVK (perovskite) $CsPbBr_3$ nanoparticles (NPs). Top view scanning electron microscope (SEM) image of PVK NPs in the inset. (b) Extinction and (c) eliminate demonstration in Pavlov's conditioning experiment (US; unconditioned stimulus, UR; unconditioned response, NS; neutral stimulus, NR; neutral response, CS; conditioned stimulus, CR; conditioned response) emulated by photonic (λ = 650 nm, P = 70 μW/cm^2, d_t = 2 s, f = 0.2 Hz) and electronic (±10 V, d_t = 2 s) spikes. (d) Punishment and (e) reward experiment to emulate the dopamine secretion behavior of a mouse. Reprinted with permission from ref 176. Copyright 2021 American Chemical Society.

layer,[149,169,170] can also induce ferroelectricity with a highest P_r of 20 μC/cm^2. Therefore, IGZO can play both the channel and capping layer roles to realize a sophisticated HfO_2-based ferroelectric synaptic transistor. In fact, a 3 × 3 device array of a ferroelectric transistor that combines HZO and IGZO has been proposed for a reconfigurable in-memory computing systems that can dual-function as logic and memory.[175]

4. ADVANCED APPLICATIONS

Emotional State and Environment Simulation. Peng et al. demonstrated the learning-to-forgetting behavior of humans under different emotional states and the environmental conditions through an IGZO synaptic transistor employing chitosan as an EDL gate dielectric (Figure 8a).[105] Note that only a voltage spike was applied to the gate electrode even though the completed transistor exhibited a synaptic plasticity upon optoelectronic stimulation. A total of 20 pulses of a positive/negative voltage spike of ±2 V (d_t = 90 ms, Δt = 100 ms) were applied to the synaptic transistor, resulting in EPSC and IPSC, respectively. In this study, the biological behavior was emulated by substituting input stimuli with devices that hardly react with voltage and wet/dry stimuli. Figure 8b shows normalized EPSC and IPSC, which represent the emotional states of uncomfortable and comfortable, and environments of quiet and noisy. The EPSC can be interpreted as uncomfortable, whereas the IPSC may represent a comfortable emotion. Wet and dry refer to the measurement conditions that are quiet and noisy, respectively. After 20 consecutive voltage spikes are removed, a time constant (t_a) of each state was defined as the half of the peak EPSC/IPSC amplitude. The higher t_a reflects that a human can memorize information longer under the quiet and comfortable conditions.

Tactile Perception. Tactile perception, the most fundamental behavior in robotics, considers the relationship between external mechanical forces and biological signal processing. Zhang et al. integrated IGZO synaptic transistor and triboelectric nanogenerator (TENG) to emulate tactile sensory behavior where the ITO bottom electrode is interconnected (Figure 8c).[107] Chitosan was also used as an EDL gate dielectric of a synaptic transistor. TENG converts an external pressure into an electrical signal output in the form of a voltage pulse (Figure 8d). With an increase in the amplitude of the pressure applied on the TENG, the output voltage spike of TENG increases accordingly in the positively upward direction (Figure 8e). The released output voltage is subsequently transported to the presynaptic voltage (gate) node of the synaptic transistor, accumulating electrons in the front channel. Accordingly, the EPSC amplitude increases when the pressure applied to TENG is coupled with a high amplitude and frequency (Figure 8f,g). This study showed that the IGZO synaptic transistor with TENG could effectively recognize a particular situation under an environment where pressure and frequencies are complexly combined. A spider's sensory capacity to identify prey was emulated under different pressure conditions, assuming a threshold EPSC to detect the prey at 400 nA. At a low amplitude with a low frequency (breeze condition) and high amplitude with low frequency (strong wind condition), the resulting EPSC does not reach the threshold such that the prey is hard to detect (Figure 8h,i). However, when the prey is captured in the spider's web, the struggling behavior to escape passes a high amplitude with a high frequency vibration, enabling the spider to sense the prey (Figure 8j).

Associative Learning. Duan et al. successfully demonstrated biological learning behaviors of the classical condition and under the reward/punishment by using optoelectronic synaptic transistor.[176] $CsPbBr_3$ perovskite nanoparticles with an optical bandgap of 2.3 eV were inserted between the top and bottom IGZO to enhance the photosynaptic plasticity of the synaptic transistor under the red light (λ = 650 nm) (Figure 9a). SiO_2 was used as a dielectric layer. Top view scanning electron microscope (SEM) image of nanoparticles is shown in the inset of Figure 9a. This is an impressive result because the IGZO synaptic transistor responds significantly to red light. For Pavlov's conditioning experiment, photonic spikes (λ = 650 nm, P = 70 μW/cm^2, d_t = 2 s, f = 0.2 Hz), negative voltage spikes (−10 V, d_t = 2 s), and PSC (threshold = 10 pA) were regarded as "food" (unconditioned stimulus, US) and "bell" (neutral stimulus, NS) to evoke a neutral response (NR) and activation of "salivation" (unconditioned

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

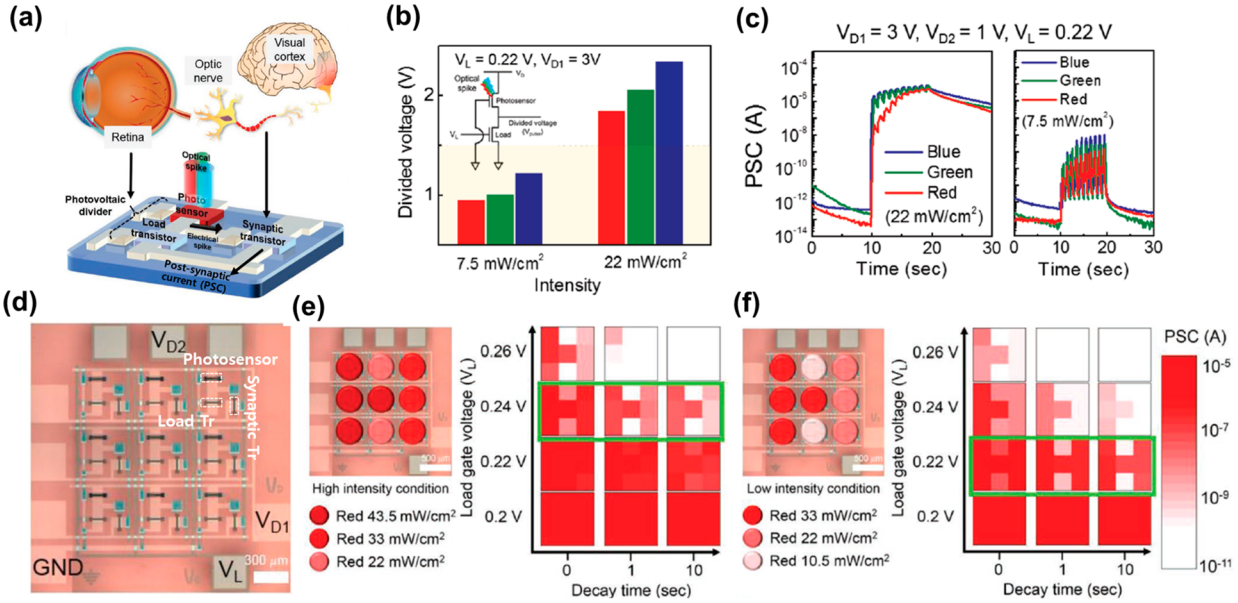


Figure 10. Light adaptation. (a) Illustration of the human visual system and a simple schematic of an optoelectronic neuromorphic system that consists of a CdSe photosensor, IGZO load and synaptic transistor. (b) Divided voltage to the gate node of a synaptic transistor when 10 photonic pulses with different conditions (P = 7.5 mW/cm^2 and 22 mW/cm^2, and λ; red = 620 nm, green = 525 nm, and blue = 465 nm) were applied to the photosensor. (c) PSC of the synaptic transistor under photonic stimulations to the photosensor. LTP (left) and STP (right) behaviors under high and low light intensity. (d) Optical microscope image of 3 × 3 device array. (e) Photopic and (f) scotopic adaptation for the "H" pattern that can be perceived when a high/low V_L is coupled with a high/low light intensity. Reprinted with permission from ref 177. Copyright 2019 Wiley-VCH.

response, UR), respectively (Figure 9b). While NS is followed by US with an interstimulus time ($\Delta t_{US\text{-}NS}$) of 5 s for 10 pairs (training), the resulting PSC exceeds the threshold. When only NS is repeated 10 times (after training), the PSC maintains the value that is excited at the initial training step, meaning NS has been converted to a conditioned stimulus (CS). When successive negative bias spikes are applied after NS has been transformed to CS, it takes a long time to initialize the PSC below the threshold (extinction). However, if one single positive bias spike (+10 V, d_t = 2 s) arrives with the same condition after 10 spikes of negative bias (CS), PSC reaches a value lower than the threshold, showing fast decaying characteristics (elimination) (Figure 9c). Another complicated "reward and punishment mechanism" was demonstrated that mirrors the dopamine secretion behavior of a mouse. When a mouse is confined in a cage, opening a window followed by providing food or nothing can be a sign of "reward" and "punishment". Photonic spikes, positive voltage spikes, and negative voltage spikes correspond to the behavior of opening a window, punishment, and reward, respectively. The threshold PSC (dopamine) was set to 15 pA. When photonic spikes are applied 10 times, PSC is close to the threshold (neutral). When subsequent positive voltage spikes (punishment) arrive followed by photonic spikes (opening a window), the PSC remains below the threshold, meaning that opening a window is no longer able to maintain the initially excited dopamine level (Figure 9d). In contrast, if photonic spikes accompany the negative voltage spikes (reward), PSC is above the threshold, indicating that opening a window can sufficiently induce a significant dopamine level (Figure 9e).

Visual Information Processing. Humans have adaptive functionalities in their visual system to perceive an object when the light intensity in the environment changes, and these are referred to as photopic and scotopic adaptation. Kwon et al. developed an optoelectronic neuromorphic system imitating a biological visual perception system by integrating simple circuits consisting of the photosensor, load, and synaptic IGZO transistors (Figure 10a).[177] The CdSe photosensor serves as a visible light (red, green, blue) absorber due to a moderate optical bandgap (~1.7 eV). The load transistor modulates the threshold of the divided voltage in order to cope with light adaptation. Note that the load and synaptic transistors used sodium-incorporated Al_2O_3 as an EDL gate dielectric. The serially connected photosensor and load transistor play the role of the artificial retina, which transforms photonic stimulation into a voltage spike. A gate electrode of the synaptic transistor receives the transformed voltage spike, acting as an artificial optic nerve. The divided voltage for the synaptic transistor input relied on the light intensity and wavelength at fixed voltage bias conditions of V_{D1} and V_L that represent the supply voltage to the photovoltaic divider and gate voltage of the load transistor, respectively (Figure 10b). Resulting PSC was measured at V_{D2} = 1 V, which is the drain voltage of the synaptic transistor. A light intensity of 7.5 mW/cm^2 was suitable to emulate the STP, and that of 22 mW/cm^2 was sufficient to imitate the LTP functions regardless of the wavelength (Figure 10c). Photopic/scotopic adaptation was implemented by a device array of 3 × 3 (Figure 10d). Consequently, a high and low intensity of red light with respective three numbers of illumination levels to describe the "H" pattern was exposed to a 3 × 3 device array. To discern the light intensities, the V_L of load transistor was employed as a threshold modulator. By measuring the PSC values as a function of the decay time under different V_L, the "H" pattern was able to be picked out when a high/low V_L is coupled with high-/low-light-intensity conditions, successfully demonstrating photopic (bright) and scotopic (dark) adaptation of the human visual system, respectively (Figure 10e,f).

Hong et al. proposed a neuromorphic active pixel image sensor array (NAPISA) of 8 × 8 to emulate the human

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

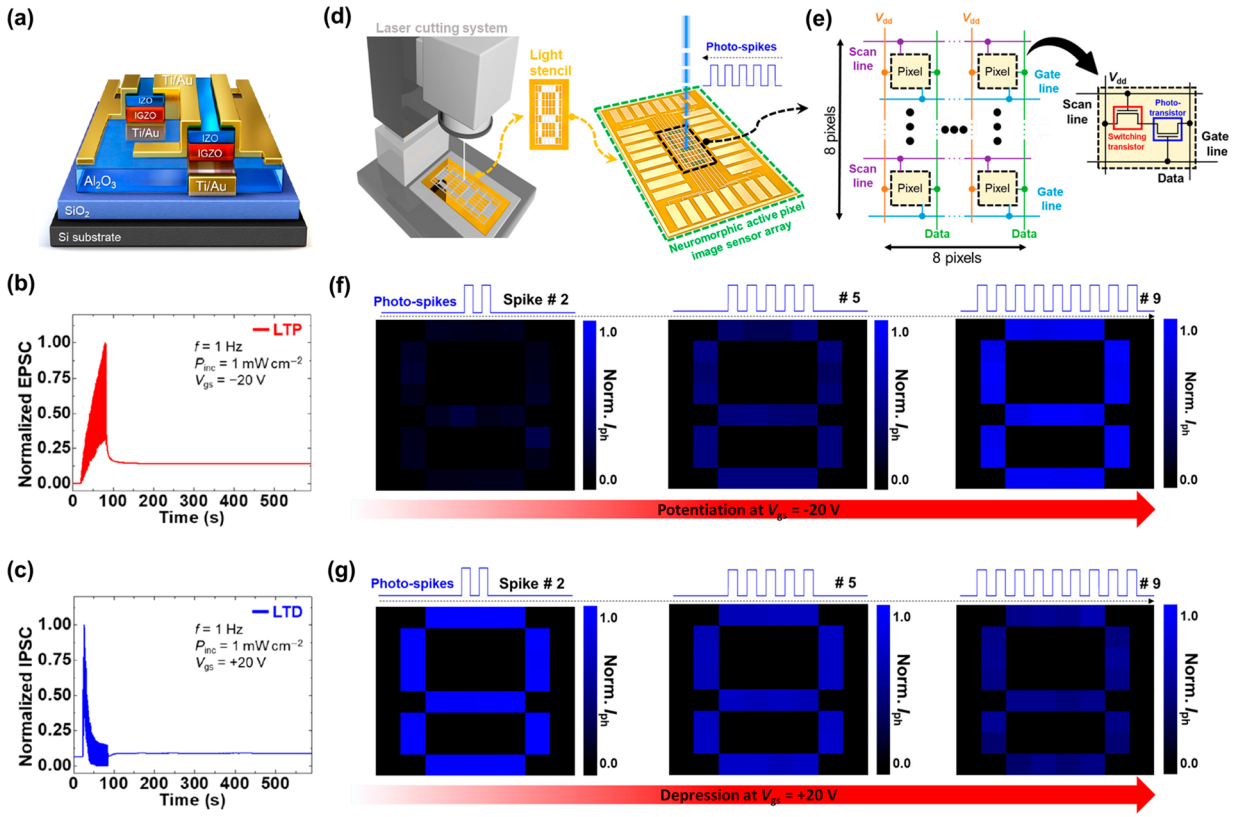


Figure 11. Visual memory. (a) Pixel configuration composed of one select transistor and one phototransistor employing IGZO/IZO heterojunctions. (b) LTP and (c) LTD characteristics under the negative and positive gate bias coupled with photonic spikes (λ = 405 nm, P = 1 mW/cm^2, f = 1 Hz). (d) Illustration to demonstrate visual memory functions by using a laser-cut Au stencil with a pattern of "8" that has 8 × 8 squares to match the same number of pixels in neuromorphic active pixel image sensory array (NAPISA). Light can only pass through the empty spaces of the stencil. (e) Equivalent circuit of NAPISA. (f) Memorizing and (g) forgetting when consecutive photonic spikes arrive the pixels under the constant negative (−20 V)/positive (+20 V) gate bias. Reprinted with permission from ref 178. Copyright 2021 American Chemical Society.

behavior of memorizing and forgetting visual information.[178] The pixel consisted of IGZO/IZO heterojunction-based select- and phototransistors (Figure 11.a) using Al_2O_3 as a dielectric layer. The phototransistor serves as an artificial synapse to exhibit synaptic plasticity. The heterojunction was formed by depositing IZO above IGZO to obtain the benefit of an intensive photoresponse from IGZO and a strong PPC behavior of IZO. The uniformity of a total of 64 selectors and phototransistors was verified. Obviously, the photonic stimulations in two terminals were utilized to trigger EPSC that leads to potentiation in Figure 3. Likewise, the multiterminal devices introduced so far used photonic stimulations with or without a negative bias to the gate electrode to emulate EPSC. However, the IPSC under photonic spikes was simply demonstrated here by applying a positive bias to the gate electrode. Consequently, LTP/LTD was emulated by applying a constant negative (−20 V)/positive bias (+20 V) coupled with successive photonic spikes (λ = 405 nm, P = 1 mW/cm^2, f = 1 Hz) to the gate electrode of the phototransistor (Figure 11b,c). To emulate the visual input, a gold thin film of a stencil with a pattern of "8" with 8 × 8 squares was developed by a laser cutting system to match the equivalent 8 × 8 pixels of NAPISA (Figure 11d,e). Then, light could only pass through the empty spaces of the stencil. Under the constant negative/positive gate bias, the pattern of "8" gradually appears and disappears after consecutive photonic spikes, successfully demonstrating "memorizing" and "forgetting", respectively (Figure 11f–g).

5. CONCLUSIONS AND PERSPECTIVES

We reviewed recent studies on IGZO-based electronic and/or photonic-synaptic transistors with various structures. Synaptic performance was confirmed to improve by combining electronic and photonic stimulation to control the PPC effect of IGZO. In a conventional TFT structure with EDL SiO_2 and Al_2O_3, versatile synaptic functions with extremely low power consumption can be emulated. Among the flash memory structures, the CTL type is expected to implement a high integration density. However, it is still unclear which program and erase mechanism is dominant for synaptic transistors between the charge storage and channel layers. Not only research to achieve a higher performance but also theoretical establishment must be simultaneously conducted. In terms of FeFET, IGZO is a highly recommended material because it can serve as a capping layer to induce the ferroelectricity of HfO_2-based materials and a channel layer at the same time. LTP characteristics could be improved by using the P_r of ferroelectric materials.

To expand the applicability of optoelectronic IGZO synaptic transistors, we suggest three initiatives. First, improve the photosynaptic plasticity in the visible range. Considering that the optical bandgap of IGZO ranges from 3.0 to 3.5 eV, recent optoelectronic IGZO synaptic transistors have been demonstrated mainly by applying UV light. If the device can exhibit discrete PSC with a wide-range spectral and intensity response, the neuromorphic system can be utilized in a great variety of applications. This can be realized via defect engineering[179] or

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

heterojunctions.[176,179,180] Second, keep the updated synaptic weight for a long time. Ferroelectricity,[163] composition engineering,[181] and heterojunctions[179,182] can improve the LTP characteristics of IGZO. Third, minimize the loss of incident light in layers other than IGZO. An antireflective thin film,[183] transparent electrode, or optimization of the device structure is required in this case.

The last section introduced various neuromorphic applications that have been demonstrated by a single artificial synapse or a circuit composed of two or more devices, utilizing various materials. A single artificial synapse can offer enhanced synaptic performance when two or more stimulations are used at the same time. Although a circuit occupies a larger area than a single synaptic device, it has an advantage in that a particular application can be effectively implemented. For example, since input stimuli can be addressed in various ways, such as with light, pressure, temperature, and humidity, designing circuits composed of appropriate sensors and synaptic transistors is the most straightforward and practical way to achieve a desired function. Therefore, research on a single synaptic device that combines multiple stimuli and circuit designs should be conducted for high integration density.

AUTHOR INFORMATION

Corresponding Author

Soo-Yeon Lee − *Department of Electrical and Computer Engineering, and Inter-university Semiconductor Research Center, Seoul National University, Seoul 08826, Republic of Korea;* orcid.org/0000-0002-9822-2570; Email: sooyeon.lee@snu.ac.kr

Authors

Yuseong Jang − *Department of Electrical and Computer Engineering, and Inter-university Semiconductor Research Center, Seoul National University, Seoul 08826, Republic of Korea;* orcid.org/0000-0002-9057-482X

Junhyeong Park − *Department of Electrical and Computer Engineering, and Inter-university Semiconductor Research Center, Seoul National University, Seoul 08826, Republic of Korea;* orcid.org/0000-0002-4318-2882

Jimin Kang − *Department of Electrical and Computer Engineering, and Inter-university Semiconductor Research Center, Seoul National University, Seoul 08826, Republic of Korea;* orcid.org/0000-0001-7263-2187

Complete contact information is available at: https://pubs.acs.org/10.1021/acsaelm.1c01088

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

This work was supported by the National Research Foundation of Korea (NRF) through the National R&D Program (No. 2021M3F3A2A01037927) and the Basic Research Laboratory (No. 2021R1A4A3032027), funded by the Korean Government (MSIT).

REFERENCES

(1) Monroe, D. Neuromorphic computing gets ready for the (really) big time. *Commun. ACM* **2014**, *57* (6), 13−15.

(2) Ho, V. M.; Lee, J.-A.; Martin, K. C. The cell biology of synaptic plasticity. *Science* **2011**, *334* (6056), 623−628.

(3) Douglas, R. J.; Martin, K. A. Recurrent neuronal circuits in the neocortex. *Current biology* **2007**, *17* (13), R496−R500.

(4) Sze, V.; Chen, Y.-H.; Emer, J.; Suleiman, A.; Zhang, Z. Hardware for machine learning: Challenges and opportunities. In *2017 IEEE Custom Integrated Circuits Conference (CICC)*; IEEE, 2017; pp 1−8.

(5) Silver, D.; Huang, A.; Maddison, C. J.; Guez, A.; Sifre, L.; van den Driessche, G.; Schrittwieser, J.; Antonoglou, I.; Panneershelvam, V.; Lanctot, M.; Dieleman, S.; Grewe, D.; Nham, J.; Kalchbrenner, N.; Sutskever, I.; Lillicrap, T.; Leach, M.; Kavukcuoglu, K.; Graepel, T.; Hassabis, D. Mastering the game of Go with deep neural networks and tree search. *Nature* **2016**, *529* (7587), 484−489.

(6) Suri, M.; Bichler, O.; Querlioz, D.; Palma, G.; Vianello, E.; Vuillaume, D.; Gamrat, C.; DeSalvo, B. CBRAM devices as binary synapses for low-power stochastic neuromorphic systems: auditory (cochlea) and visual (retina) cognitive processing applications. In *2012 International Electron Devices Meeting*; IEEE, 2012; pp 10.13.11−10.13.14.

(7) Alibart, F.; Zamanidoost, E.; Strukov, D. B. Pattern classification by memristive crossbar circuits using ex situ and in situ training. *Nat. Commun.* **2013**, *4* (1), 1−7.

(8) Lee, D.; Park, J.; Moon, K.; Jang, J.; Park, S.; Chu, M.; Kim, J.; Noh, J.; Jeon, M.; Lee, B. H. Oxide based nanoscale analog synapse device for neural signal recognition system. In *2015 IEEE International Electron Devices Meeting (IEDM)*; IEEE, 2015; pp 4.7.1−4.7.4.

(9) Wang, C.; Liang, S.-J.; Wang, C.-Y.; Yang, Z.-Z.; Ge, Y.; Pan, C.; Shen, X.; Wei, W.; Zhao, Y.; Zhang, Z.; Cheng, B.; Zhang, C.; Miao, F. Scalable massively parallel computing using continuous-time data representation in nanoscale crossbar array. *Nat. Nanotechnol.* **2021**, *16*, 1079−1085.

(10) Li, C.; Wang, Z.; Rao, M.; Belkin, D.; Song, W.; Jiang, H.; Yan, P.; Li, Y.; Lin, P.; Hu, M.; Ge, N.; Strachan, J. P.; Barnell, M.; Wu, Q.; Williams, R. S.; Yang, J. J.; Xia, Q. Long short-term memory networks in memristor crossbar arrays. *Nature Machine Intelligence* **2019**, *1* (1), 49−57.

(11) Eryilmaz, S. B.; Kuzum, D.; Jeyasingh, R. G.; Kim, S.; BrightSky, M.; Lam, C.; Wong, H.-S. P. Experimental demonstration of array-level learning with phase change synaptic devices. In *2013 IEEE International Electron Devices Meeting*; IEEE, 2013; pp 25.25. 21−25.25. 24.

(12) Seo, J.-S.; Brezzo, B.; Liu, Y.; Parker, B. D.; Esser, S. K.; Montoye, R. K.; Rajendran, B.; Tierno, J. A.; Chang, L.; Modha, D. S. A 45nm CMOS neuromorphic chip with a scalable architecture for learning in networks of spiking neurons. In *2011 IEEE Custom Integrated Circuits Conference (CICC)*, 2011; IEEE: pp 1−4.

(13) Han, H.; Yu, H.; Wei, H.; Gong, J.; Xu, W. Recent progress in three-terminal artificial synapses: from device to system. *Small* **2019**, *15* (32), 1900695.

(14) Nishitani, Y.; Kaneko, Y.; Ueda, M.; Morie, T.; Fujii, E. Three-terminal ferroelectric synapse device with concurrent learning function for artificial neural networks. *J. Appl. Phys.* **2012**, *111* (12), 124108.

(15) Zhou, T.; Lin, X.; Wu, J.; Chen, Y.; Xie, H.; Li, Y.; Fan, J.; Wu, H.; Fang, L.; Dai, Q. Large-scale neuromorphic optoelectronic computing with a reconfigurable diffractive processing unit. *Nat. Photonics* **2021**, *15* (5), 367−373.

(16) Wuttig, M.; Bhaskaran, H.; Taubner, T. Phase-change materials for non-volatile photonic applications. *Nat. Photonics* **2017**, *11* (8), 465−476.

(17) Ríos, C.; Youngblood, N.; Cheng, Z.; Le Gallo, M.; Pernice, W. H.; Wright, C. D.; Sebastian, A.; Bhaskaran, H. In-memory computing on a photonic platform. *Science advances* **2019**, *5* (2), No. eaau5759.

(18) Feldmann, J.; Youngblood, N.; Karpov, M.; Gehring, H.; Li, X.; Stappers, M.; Le Gallo, M.; Fu, X.; Lukashchuk, A.; Raja, A. S.; Liu, J.; Wright, C. D.; Sebastian, A.; Kippenberg, T. J.; Pernice, W. H. P.; Bhaskaran, H. Parallel convolutional processing using an integrated photonic tensor core. *Nature* **2021**, *589* (7840), 52−58.

(19) Peng, H.-T.; Nahmias, M. A.; De Lima, T. F.; Tait, A. N.; Shastri, B. J. Neuromorphic photonic integrated circuits. *IEEE J. Sel. Top. Quantum Electron.* **2018**, *24* (6), 1−15.

(20) Shastri, B. J.; Tait, A. N.; de Lima, T. F.; Pernice, W. H.; Bhaskaran, H.; Wright, C. D.; Prucnal, P. R. Photonics for artificial intelligence and neuromorphic computing. *Nat. Photonics* **2021**, *15* (2), 102−114.
(21) Yin, L.; Han, C.; Zhang, Q.; Ni, Z.; Zhao, S.; Wang, K.; Li, D.; Xu, M.; Wu, H.; Pi, X.; Yang, D. Synaptic silicon-nanocrystal phototransistors for neuromorphic computing. *Nano Energy* **2019**, *63*, 103859.
(22) Luo, Z.-D.; Xia, X.; Yang, M.-M.; Wilson, N. R.; Gruverman, A.; Alexe, M. Artificial optoelectronic synapses based on ferroelectric field-effect enabled 2D transition metal dichalcogenide memristive transistors. *ACS Nano* **2020**, *14* (1), 746−754.
(23) Ma, F.; Zhu, Y.; Xu, Z.; Liu, Y.; Zheng, X.; Ju, S.; Li, Q.; Ni, Z.; Hu, H.; Chai, Y.; Wu, C.; Kim, T. W.; Li, F. Optoelectronic perovskite synapses for neuromorphic computing. *Adv. Funct. Mater.* **2020**, *30* (11), 1908901.
(24) Qian, C.; Oh, S.; Choi, Y.; Kim, J.-H.; Sun, J.; Huang, H.; Yang, J.; Gao, Y.; Park, J.-H.; Cho, J. H. Solar-stimulated optoelectronic synapse based on organic heterojunction with linearly potentiated synaptic weight for neuromorphic computing. *Nano Energy* **2019**, *66*, 104095.
(25) Zhu, R.; Liang, H.; Hu, S.; Wang, Y.; Mei, Z. Amorphous-Ga2O3 Optoelectronic Synapses with Ultra-low Energy Consumption. *Advanced Electronic Materials* **2022**, *8*, 2100741.
(26) Cho, S. W.; Kwon, S. M.; Lee, M.; Jo, J.-W.; Heo, J. S.; Kim, Y.-H.; Cho, H. K.; Park, S. K. Multi-spectral gate-triggered heterogeneous photonic neuro-transistors for power-efficient brain-inspired neuromorphic computing. *Nano Energy* **2019**, *66*, 104097.
(27) Duan, H.; Javaid, K.; Liang, L.; Huang, L.; Yu, J.; Zhang, H.; Gao, J.; Zhuge, F.; Chang, T.-C.; Cao, H. Broadband Optoelectronic Synaptic Thin-Film Transistors Based on Oxide Semiconductors. *Physica status solidi (RRL)−Rapid Research Letters* **2020**, *14* (4), 1900630.
(28) Dai, S.; Zhao, Y.; Wang, Y.; Zhang, J.; Fang, L.; Jin, S.; Shao, Y.; Huang, J. Recent advances in transistor-based artificial synapses. *Adv. Funct. Mater.* **2019**, *29* (42), 1903700.
(29) Huang, H.; Bodnarchuk, M. I.; Kershaw, S. V.; Kovalenko, M. V.; Rogach, A. L. Lead halide perovskite nanocrystals in the research spotlight: stability and defect tolerance. *ACS energy letters* **2017**, *2* (9), 2071−2083.
(30) Choi, J.; Han, J. S.; Hong, K.; Kim, S. Y.; Jang, H. W. Organic−inorganic hybrid halide perovskites for memories, transistors, and artificial synapses. *Adv. Mater.* **2018**, *30* (42), 1704002.
(31) Kim, D.; Jung, H. J.; Park, I. J.; Larson, B. W.; Dunfield, S. P.; Xiao, C.; Kim, J.; Tong, J.; Boonmongkolras, P.; Ji, S. G.; Zhang, F.; Pae, S. R.; Kim, M.; Kang, S. B.; Dravid, V.; Berry, J.; Kim, J. Y.; Zhu, K.; Kim, D. H.; Shin, B. Efficient, stable silicon tandem cells enabled by anion-engineered wide-bandgap perovskites. *Science* **2020**, *368* (6487), 155−160.
(32) Shin, H. J.; Takasugi, S.; Park, K. M.; Choi, S. H.; Jeong, Y. S.; Song, B. C.; Kim, H. S.; Oh, C. H.; Ahn, B. C. 7.1: Invited Paper: Novel OLED display technologies for large-size UHD OLED TVs. In *SID Symposium Digest of Technical Papers*; Wiley Online Library, 2015; Vol. *46*, pp 53−56.
(33) Kamiya, T.; Nomura, K.; Hosono, H. Origins of high mobility and low operation voltage of amorphous oxide TFTs: Electronic structure, electron transport, defects and doping. *Journal of display Technology* **2009**, *5* (12), 468−483.
(34) Jang, J. T.; Park, J.; Ahn, B. D.; Kim, D. M.; Choi, S.-J.; Kim, H.-S.; Kim, D. H. Study on the photoresponse of amorphous In−Ga−Zn−O and zinc oxynitride semiconductor devices by the extraction of sub-gap-state distribution and device simulation. *ACS Appl. Mater. Interfaces* **2015**, *7* (28), 15570−15577.
(35) Jeon, S.; Ahn, S.-E.; Song, I.; Kim, C. J.; Chung, U-I.; Lee, E.; Yoo, I.; Nathan, A.; Lee, S.; Ghaffarzadeh, K.; Robertson, J.; Kim, K. Gated three-terminal device architecture to eliminate persistent photoconductivity in oxide semiconductor photosensor arrays. *Nature materials* **2012**, *11* (4), 301−305.
(36) Seo, S.; Jo, S.-H.; Kim, S.; Shim, J.; Oh, S.; Kim, J.-H.; Heo, K.; Choi, J.-W.; Choi, C.; Oh, S.; Kuzum, D.; Wong, H.-S. P.; Park, J.-H. Artificial optic-neural synapse for colored and color-mixed pattern recognition. *Nat. Commun.* **2018**, *9* (1), 1−8.
(37) Lee, Y.; Oh, J. Y.; Xu, W.; Kim, O.; Kim, T. R.; Kang, J.; Kim, Y.; Son, D.; Tok, J. B.-H.; Park, M. J.; Bao, Z.; Lee, T.-W. Stretchable organic optoelectronic sensorimotor synapse. *Science advances* **2018**, *4* (11), No. eaat7387.
(38) Kandel, E. R. Nerve cells and behavior. In *Principles of Neural Science*; McGraw-Hill, 2000; pp 18−32.
(39) Burkitt, A. N. A review of the integrate-and-fire neuron model: I. Homogeneous synaptic input. *Biological cybernetics* **2006**, *95* (1), 1−19.
(40) Liu, Y.-H.; Wang, X.-J. Spike-frequency adaptation of a generalized leaky integrate-and-fire model neuron. *Journal of computational neuroscience* **2001**, *10* (1), 25−45.
(41) Dutta, S.; Kumar, V.; Shukla, A.; Mohapatra, N. R.; Ganguly, U. Leaky integrate and fire neuron by charge-discharge dynamics in floating-body MOSFET. *Sci. Rep.* **2017**, *7* (1), 1−7.
(42) Jiang, Y.; Huang, P.; Zhu, D.; Zhou, Z.; Han, R.; Liu, L.; Liu, X.; Kang, J. Design and hardware implementation of neuromorphic systems with RRAM synapses and threshold-controlled neurons for pattern recognition. *IEEE Transactions on Circuits and Systems I: Regular Papers* **2018**, *65* (9), 2726−2738.
(43) Abbott, L. F.; Nelson, S. B. Synaptic plasticity: taming the beast. *Nature neuroscience* **2000**, *3* (11), 1178−1183.
(44) Bliss, T. V.; Collingridge, G. L. A synaptic model of memory: long-term potentiation in the hippocampus. *Nature* **1993**, *361* (6407), 31−39.
(45) Buonomano, D. V. Decoding temporal information: a model based on short-term synaptic plasticity. *J. Neurosci.* **2000**, *20* (3), 1129−1141.
(46) Kuzum, D.; Yu, S.; Wong, H. P. Synaptic electronics: materials, devices and applications. *Nanotechnology* **2013**, *24* (38), 382001.
(47) Kim, D. W.; Yang, J. C.; Lee, S.; Park, S. Neuromorphic processing of pressure signal using integrated sensor-synaptic device capable of selective and reversible short-and long-term plasticity operation. *ACS Appl. Mater. Interfaces* **2020**, *12* (20), 23207−23216.
(48) Li, E.; Lin, W.; Yan, Y.; Yang, H.; Wang, X.; Chen, Q.; Lv, D.; Chen, G.; Chen, H.; Guo, T. Synaptic transistor capable of accelerated learning induced by temperature-facilitated modulation of synaptic plasticity. *ACS Appl. Mater. Interfaces* **2019**, *11* (49), 46008−46016.
(49) Zucker, R. S.; Regehr, W. G. Short-term synaptic plasticity. *Annual review of physiology* **2002**, *64* (1), 355−405.
(50) Kwon, S. M.; Kwak, J. Y.; Song, S.; Kim, J.; Jo, C.; Cho, S. S.; Nam, S.-J.; Kim, J.; Park, G.-S.; Kim, Y.-H.; Park, S. K. Large-Area Pixelized Optoelectronic Neuromorphic Devices with Multispectral Light-Modulated Bidirectional Synaptic Circuits. *Adv. Mater.* **2021**, *33*, 2105017.
(51) Caporale, N.; Dan, Y. Spike timing−dependent plasticity: a Hebbian learning rule. *Annu. Rev. Neurosci.* **2008**, *31*, 25−46.
(52) Bi, G.-Q.; Poo, M.-M. Synaptic modifications in cultured hippocampal neurons: dependence on spike timing, synaptic strength, and postsynaptic cell type. *J. Neurosci.* **1998**, *18* (24), 10464−10472.
(53) Rachmuth, G.; Shouval, H. Z.; Bear, M. F.; Poon, C.-S. A biophysically-based neuromorphic model of spike rate-and timing-dependent plasticity. *Proc. Natl. Acad. Sci. U. S. A.* **2011**, *108* (49), E1266−E1274.
(54) Sjöström, P. J.; Turrigiano, G. G.; Nelson, S. B. Rate, timing, and cooperativity jointly determine cortical synaptic plasticity. *Neuron* **2001**, *32* (6), 1149−1164.
(55) Feldman, D. E. The spike-timing dependence of plasticity. *Neuron* **2012**, *75* (4), 556−571.
(56) Moon, K.; Lim, S.; Park, J.; Sung, C.; Oh, S.; Woo, J.; Lee, J.; Hwang, H. RRAM-based synapse devices for neuromorphic systems. *Faraday Discuss.* **2019**, *213*, 421−451.
(57) Wang, C.; Li, Y.; Wang, Y.; Xu, X.; Fu, M.; Liu, Y.; Lin, Z.; Ling, H.; Gkoupidenis, P.; Yi, M.; Xie, L.; Yan, F.; Huang, W. Thin-

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

film transistors for emerging neuromorphic electronics: fundamentals, materials, and pattern recognition. *Journal of Materials Chemistry C* **2021**, *9* (35), 11464−11483.
(58) Sun, J.; Oh, S.; Choi, Y.; Seo, S.; Oh, M. J.; Lee, M.; Lee, W. B.; Yoo, P. J.; Cho, J. H.; Park, J. H. Optoelectronic synapse based on igzo-alkylated graphene oxide hybrid structure. *Adv. Funct. Mater.* **2018**, *28* (47), 1804397.
(59) Son, K.-S.; Jung, J.-S.; Lee, K.-H.; Kim, T.-S.; Park, J.-S.; Choi, Y.-H.; Park, K.; Kwon, J.-Y.; Koo, B.; Lee, S.-Y. Characteristics of double-gate Ga−In−Zn−O thin-film transistor. *IEEE Electron Device Lett.* **2010**, *31* (3), 219−221.
(60) Lim, H.; Yin, H.; Park, J.-S.; Song, I.; Kim, C.; Park, J.; Kim, S.; Kim, S.-W.; Lee, C. B.; Kim, Y. C.; Park, Y. S.; Kang, D. Double gate GaInZnO thin film transistors. *Applied physics letters* **2008**, *93* (6), 063505.
(61) Baek, G.; Abe, K.; Kuo, A.; Kumomi, H.; Kanicki, J. Electrical properties and stability of dual-gate coplanar homojunction DC sputtered amorphous indium−gallium−zinc−oxide thin-film transistors and its application to AM-OLEDs. *IEEE Transactions on electron devices* **2011**, *58* (12), 4344−4353.
(62) Ahn, M.-J.; Cho, W.-J. Transparent multi-level-cell nonvolatile memory with dual-gate amorphous indium-gallium-zinc oxide thin-film transistors. *Appl. Phys. Lett.* **2016**, *109* (25), 252106.
(63) Hays, D. C.; Gila, B.; Pearton, S.; Ren, F. Energy band offsets of dielectrics on InGaZnO4. *Applied Physics Reviews* **2017**, *4* (2), 021301.
(64) Nomura, K.; Ohta, H.; Takagi, A.; Kamiya, T.; Hirano, M.; Hosono, H. Room-temperature fabrication of transparent flexible thin-film transistors using amorphous oxide semiconductors. *Nature* **2004**, *432* (7016), 488−492.
(65) Yabuta, H.; Sano, M.; Abe, K.; Aiba, T.; Den, T.; Kumomi, H.; Nomura, K.; Kamiya, T.; Hosono, H. High-mobility thin-film transistor with amorphous In Ga Zn O 4 channel fabricated by room temperature rf-magnetron sputtering. *Applied physics letters* **2006**, *89* (11), 112123.
(66) Jiang, J.; Wan, Q.; Sun, J.; Lu, A. Ultralow-voltage transparent electric-double-layer thin-film transistors processed at room-temperature. *Appl. Phys. Lett.* **2009**, *95* (15), 152114.
(67) Lee, M.; Lee, W.; Choi, S.; Jo, J. W.; Kim, J.; Park, S. K.; Kim, Y. H. Brain-inspired photonic neuromorphic devices using photodynamic amorphous oxide semiconductors and their persistent photoconductivity. *Adv. Mater.* **2017**, *29* (28), 1700951.
(68) Hosono, H. Ionic amorphous oxide semiconductors: Material design, carrier transport, and device application. *Journal of non-crystalline solids* **2006**, *352* (9−20), 851−858.
(69) Fortunato, E.; Barquinha, P.; Martins, R. Oxide semiconductor thin-film transistors: a review of recent advances. *Advanced materials* **2012**, *24* (22), 2945−2986.
(70) Noh, H.-K.; Chang, K.-J.; Ryu, B.; Lee, W.-J. Electronic structure of oxygen-vacancy defects in amorphous In-Ga-Zn-O semiconductors. *Phys. Rev. B* **2011**, *84* (11), 115205.
(71) Kim, H. Y.; Park, Y. J.; Kim, J.; Han, C. J.; Lee, J.; Kim, Y.; Greco, T.; Ippen, C.; Wedel, A.; Ju, B.-K.; Oh, M. S. Transparent InP Quantum Dot Light-Emitting Diodes with ZrO2 Electron Transport Layer and Indium Zinc Oxide Top Electrode. *Adv. Funct. Mater.* **2016**, *26* (20), 3454−3461.
(72) Chung, J.; Tak, Y. J.; Kim, W.-G.; Kang, B. H.; Kim, H. J. Artificially Fabricated Subgap States for Visible-Light Absorption in Indium−Gallium−Zinc Oxide Phototransistor with Solution-Processed Oxide Absorption Layer. *ACS Appl. Mater. Interfaces* **2019**, *11* (42), 38964−38972.
(73) Ide, K.; Nomura, K.; Hosono, H.; Kamiya, T. Electronic Defects in Amorphous Oxide Semiconductors: A Review. *Physica status solidi (a)* **2019**, *216* (5), 1800372.
(74) Li, H.; Guo, Y.; Robertson, J. Oxygen vacancies and hydrogen in amorphous In-Ga-Zn-O and ZnO. *Physical Review Materials* **2018**, *2* (7), 074601.
(75) Mativenga, M.; Haque, F.; Billah, M. M.; Um, J. G. Origin of light instability in amorphous IGZO thin-film transistors and its suppression. *Sci. Rep.* **2021**, *11* (1), 14618.
(76) Kang, B. H.; Kim, W.-G.; Chung, J.; Lee, J. H.; Kim, H. J. Simple hydrogen plasma doping process of amorphous indium gallium zinc oxide-based phototransistors for visible light detection. *ACS Appl. Mater. Interfaces* **2018**, *10* (8), 7223−7230.
(77) Janotti, A.; Van de Walle, C. G. Oxygen vacancies in ZnO. *Appl. Phys. Lett.* **2005**, *87* (12), 122102.
(78) Lee, M.; Kim, M.; Jo, J.-W.; Park, S. K.; Kim, Y.-H. Suppression of persistent photo-conductance in solution-processed amorphous oxide thin-film transistors. *Appl. Phys. Lett.* **2018**, *112* (5), 052103.
(79) Liu, P. T.; Ruan, D. B.; Yeh, X. Y.; Chiu, Y. C.; Zheng, G. T.; Sze, S. M. Highly responsive blue light sensor with amorphous indium-zinc-oxide thin-film transistor based architecture. *Sci. Rep.* **2018**, *8* (1), 1−9.
(80) Yoo, H.; Lee, I. S.; Jung, S.; Rho, S. M.; Kang, B. H.; Kim, H. J. A Review of Phototransistors Using Metal Oxide Semiconductors: Research Progress and Future Directions. *Adv. Mater.* **2021**, *33*, 2006091.
(81) Jung, C.; Kim, S. M.; Moon, H.; Han, G.; Kwon, J.; Hong, Y. K.; Omkaram, I.; Yoon, Y.; Kim, S.; Park, J. Highly crystalline CVD-grown multilayer MoSe 2 thin film transistor for fast photodetector. *Sci. Rep.* **2015**, *5* (1), 1−9.
(82) Pawbake, A. S.; Waykar, R. G.; Late, D. J.; Jadkar, S. R. Highly transparent wafer-scale synthesis of crystalline WS2 nanoparticle thin film for photodetector and humidity-sensing applications. *ACS Appl. Mater. Interfaces* **2016**, *8* (5), 3359−3365.
(83) Wang, J.; Chen, Y.; Kong, L.-A.; Fu, Y.; Gao, Y.; Sun, J. Deep-ultraviolet-triggered neuromorphic functions in In-Zn-O phototransistors. *Appl. Phys. Lett.* **2018**, *113* (15), 151101.
(84) Hu, L.; Yang, J.; Wang, J.; Cheng, P.; Chua, L. O.; Zhuge, F. All-optically controlled memristor for optoelectronic neuromorphic computing. *Adv. Funct. Mater.* **2021**, *31* (4), 2005582.
(85) Duan, N.; Li, Y.; Chiang, H.-C.; Chen, J.; Pan, W.-Q.; Zhou, Y.-X.; Chien, Y.-C.; He, Y.-H.; Xue, K.-H.; Liu, G.; Chang, T.-C.; Miao, X.-S. An electro-photo-sensitive synaptic transistor for edge neuromorphic visual systems. *Nanoscale* **2019**, *11* (38), 17590−17599.
(86) Kim, S.; Lim, M.; Kim, Y.; Kim, H.-D.; Choi, S.-J. Impact of synaptic device variations on pattern recognition accuracy in a hardware neural network. *Sci. Rep.* **2018**, *8*, 2638.
(87) Kim, S.; Kim, H.-D.; Choi, S.-J. Impact of synaptic device variations on classification accuracy in a binarized neural network. *Sci. Rep.* **2019**, *9* (1), 1−7.
(88) Kim, S.; Lee, Y.; Kim, H.-D.; Choi, S.-J. Precision-extension technique for accurate vector−matrix multiplication with a CNT transistor crossbar array. *Nanoscale* **2019**, *11* (44), 21449−21457.
(89) Kim, S. K.; Jeong, Y.; Bidenko, P.; Lim, H.-R.; Jeon, Y.-R.; Kim, H.; Lee, Y. J.; Geum, D.-M.; Han, J.; Choi, C.; Kim, H.-j.; Kim, S. 3D stackable synaptic transistor for 3D integrated artificial neural networks. *ACS Appl. Mater. Interfaces* **2020**, *12* (6), 7372−7380.
(90) Park, S.; Jang, J. T.; Hwang, Y.; Lee, H.; Choi, W. S.; Kang, D.; Kim, C.; Kim, H.; Kim, D. H. Effect of the Gate Dielectric Layer of Flexible InGaZnO Synaptic Thin-Film Transistors on Learning Behavior. *ACS Applied Electronic Materials* **2021**, *3* (9), 3972−3979.
(91) Guerra-Nuñez, C.; Döbeli, M.; Michler, J.; Utke, I. Reaction and growth mechanisms in Al2O3 deposited via atomic layer deposition: elucidating the hydrogen source. *Chem. Mater.* **2017**, *29* (20), 8690−8703.
(92) Cho, S.-I.; Jeon, J. B.; Kim, J. H.; Lee, S. H.; Jeong, W.; Kim, J.; Kim, G.; Kim, K. M.; Park, S.-H. K. Synaptic transistors with human brain-like fJ energy consumption via double oxide semiconductor engineering for neuromorphic electronics. *Journal of Materials Chemistry C* **2021**, *9* (32), 10243−10253.
(93) Zhou, J.; Wan, C.; Zhu, L.; Shi, Y.; Wan, Q. Synaptic behaviors mimicked in flexible oxide-based transistors on plastic substrates. *IEEE Electron Device Lett.* **2013**, *34* (11), 1433−1435.

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

(94) Du, H.; Lin, X.; Xu, Z.; Chu, D. Electric double-layer transistors: a review of recent progress. *J. Mater. Sci.* **2015**, *50* (17), 5641−5673.
(95) He, Y.; Yang, Y.; Nie, S.; Liu, R.; Wan, Q. Electric-double-layer transistors for synaptic devices and neuromorphic systems. *Journal of Materials Chemistry C* **2018**, *6* (20), 5336−5352.
(96) Ling, H.; Koutsouras, D. A.; Kazemzadeh, S.; Van De Burgt, Y.; Yan, F.; Gkoupidenis, P. Electrolyte-gated transistors for synaptic electronics, neuromorphic computing, and adaptable biointerfacing. *Applied Physics Reviews* **2020**, *7* (1), 011307.
(97) Yang, X.; Liu, Z.; Luo, J.; Zhang, C.; He, Y.; Shi, Y.; Wan, Q. Freestanding Multi-Gate Amorphous Oxide-Based TFTs on Graphene Oxide Enhanced Electrolyte Membranes. *IEEE Electron Device Lett.* **2020**, *41* (9), 1360−1363.
(98) Ma, X.; Zhang, J.; Cai, W.; Wang, H.; Wilson, J.; Wang, Q.; Xin, Q.; Song, A. A sputtered silicon oxide electrolyte for high-performance thin-film transistors. *Sci. Rep.* **2017**, *7* (1), 1−6.
(99) He, Y.; Nie, S.; Liu, R.; Shi, Y.; Wan, Q. Indium−gallium−zinc−oxide Schottky synaptic transistors for silent synapse conversion emulation. *IEEE Electron Device Lett.* **2019**, *40* (1), 139−142.
(100) Yang, Y.; He, Y.; Nie, S.; Shi, Y.; Wan, Q. Light stimulated IGZO-based electric-double-layer transistors for photoelectric neuromorphic devices. *IEEE Electron Device Lett.* **2018**, *39* (6), 897−900.
(101) Park, Y.; Kim, M.-K.; Lee, J.-S. Artificial synaptic transistors based on Schottky barrier height modulation using reduced graphene oxides. *Carbon* **2020**, *165*, 455−460.
(102) Fu, Y. M.; Zhang, J.; Cai, W.; Wilson, J.; Brownless, J.; Wei, T.; Song, A. Sputtered Oxide Thin-Film Transistors With Tunable Synaptic Spiking Behavior at 1 V. *IEEE Trans. Electron Devices* **2021**, *68* (6), 2736−2741.
(103) Guo, L. Q.; Zhu, L. Q.; Ding, J. N.; Huang, Y. K. Paired-pulse facilitation achieved in protonic/electronic hybrid indium gallium zinc oxide synaptic transistors. *AIP Advances* **2015**, *5* (8), 087112.
(104) Jin, D.-G.; Kim, S.-H.; Kim, S.-G.; Park, J.; Park, E.; Yu, H.-Y. Enhancement of Synaptic Characteristics Achieved by the Optimization of Proton−Electron Coupling Effect in a Solid-State Electrolyte-Gated Transistor. *Small* **2021**, *17* (30), 2100242.
(105) Peng, C.; Jiang, W.; Li, Y.; Li, X.; Zhang, J. Photoelectric IGZO Electric-Double-Layer Transparent Artificial Synapses for Emotional State Simulation. *ACS Applied Electronic Materials* **2019**, *1* (11), 2406−2414.
(106) Kwon, S. M.; Cho, S. W.; Kim, M.; Heo, J. S.; Kim, Y. H.; Park, S. K. Environment-Adaptable Artificial Visual Perception Behaviors Using a Light-Adjustable Optoelectronic Neuromorphic Device Array. *Adv. Mater.* **2019**, *31* (52), 1906433.
(107) Zhang, C.; Li, S.; He, Y.; Chen, C.; Jiang, S.; Yang, X.; Wang, X.; Pan, L.; Wan, Q. Oxide synaptic transistors coupled with triboelectric nanogenerators for bio-inspired tactile sensing application. *IEEE Electron Device Lett.* **2020**, *41* (4), 617−620.
(108) Bates, J. B.; Wang, J. C.; Perkins, R. A. Mechanisms for hydrogen diffusion in TiO2. *Phys. Rev. B* **1979**, *19* (8), 4130−4139.
(109) Hughart, D. R.; Schrimpf, R. D.; Fleetwood, D. M.; Rowsey, N. L.; Law, M. E.; Tuttle, B. R.; Pantelides, S. T. The Effects of Proton-Defect Interactions on Radiation-Induced Interface-Trap Formation and Annealing. *IEEE Trans. Nucl. Sci.* **2012**, *59* (6), 3087−3092.
(110) Park, J.; Kim, H.; Choi, P.; Jeon, B.; Lee, J.; Oh, C.; Kim, B.; Choi, B. Effect of ALD- and PEALD- Grown Al2O3 Gate Insulators on Electrical and Stability Properties for a-IGZO Thin-Film Transistor. *Electronic Materials Letters* **2021**, *17* (4), 299−306.
(111) Ding, X.; Zhang, H.; Zhang, J.; Li, J.; Shi, W.; Jiang, X.; Zhang, Z. IGZO thin film transistors with Al2O3 gate insulators fabricated at different temperatures. *Materials Science in Semiconductor Processing* **2015**, *29*, 69−75.
(112) Cai, W.; Zhang, J.; Wilson, J.; Ma, X.; Wang, H.; Zhang, X.; Xin, Q.; Song, A. Oxide-Based Electric-Double-Layer Thin-Film Transistors on a Flexible Substrate. *IEEE Electron Device Lett.* **2017**, *38* (12), 1680−1683.
(113) Wan, C. J.; Zhu, L. Q.; Zhou, J. M.; Shi, Y.; Wan, Q. Memory and learning behaviors mimicked in nanogranular SiO2-based proton conductor gated oxide-based synaptic transistors. *Nanoscale* **2013**, *5* (21), 10194−10199.
(114) Zhang, H.; Guo, L.; Wan, Q. Nanogranular Al2O3 proton conducting films for low-voltage oxide-based homojunction thin-film transistors. *Journal of Materials Chemistry C* **2013**, *1* (15), 2781−2786.
(115) Wei, L.; Huang, W.; Fang, X.; Wang, X.; Mou, P.; Shao, F.; Gu, X. Humidity Stability of All-Sputtered Metal-Oxide Electric-Double-Layer Transistors. *IEEE Trans. Electron Devices* **2020**, *67* (12), 5532−5536.
(116) Shi, Q.; Liu, D.; Hao, D.; Zhang, J.; Tian, L.; Xiong, L.; Huang, J. Printable, ultralow-power ternary synaptic transistors for multifunctional information processing system. *Nano Energy* **2021**, *87*, 106197.
(117) Zhao, R.; Xie, L.; Zhuang, R.; Wu, T.; Zhao, R.; Wang, L.; Sun, L.; Hua, Y. Interfacial Defect Passivation and Charge Carrier Management for Efficient Perovskite Solar Cells via a Highly Crystalline Small Molecule. *ACS Energy Letters* **2021**, *6* (12), 4209−4219.
(118) Haase, K.; Teixeira da Rocha, C.; Hauenstein, C.; Zheng, Y.; Hambsch, M.; Mannsfeld, S. C. B. High-Mobility, Solution-Processed Organic Field-Effect Transistors from C8-BTBT:Polystyrene Blends. *Advanced Electronic Materials* **2018**, *4* (8), 1800076.
(119) Citri, A.; Malenka, R. C. Synaptic plasticity: multiple forms, functions, and mechanisms. *Neuropsychopharmacology* **2008**, *33* (1), 18−41.
(120) Tai, M. C.; Wang, Y. X.; Chang, T. C.; Lin, C. C.; Tu, Y. F.; Hung, Y. H.; Ciou, F. M.; Lin, Y. S.; Sze, S. Heterojunction Channels in Oxide Semiconductors for Visible-Blind Nonvolatile Optoelectronic Memories. *Advanced Electronic Materials* **2020**, *6* (11), 2000747.
(121) Jeong, D. Y.; Billah, M. M.; Jang, J. Amorphous InGaZnO/Poly-Si Coplanar Heterojunction TFT for Memory Applications. *IEEE Electron Device Lett.* **2021**, *42* (8), 1172−1175.
(122) Zhang, C.; Li, D.; Lai, P.; Huang, X. An InGaZnO Charge-Trapping Nonvolatile Memory With The Same Structure of a Thin-Film Transistor. *IEEE Electron Device Lett.* **2022**, *43*, 1−1.
(123) Bez, R.; Pirovano, A. Overview of non-volatile memory technology: markets, technologies and trends. In *Advances in Non-volatile Memory and Storage Technology*; Elsevier, 2014; pp 1−24.
(124) Naqi, M.; Kwon, N.; Jung, S. H.; Pujar, P.; Cho, H. W.; Cho, Y. I.; Cho, H. K.; Lim, B.; Kim, S. High-Performance Non-Volatile InGaZnO Based Flash Memory Device Embedded with a Monolayer Au Nanoparticles. *Nanomaterials* **2021**, *11* (5), 1101.
(125) Qian, S.-B.; Shao, Y.; Liu, W.-J.; Zhang, D. W.; Ding, S.-J. Erasing-modes dependent performance of a-IGZO TFT memory with atomic-layer-deposited Ni nanocrystal charge storage layer. *IEEE Trans. Electron Devices* **2017**, *64* (7), 3023−3027.
(126) Kim, J.; Kim, Y.; Kwon, O.; Kim, T.; Oh, S.; Jin, S.; Park, W.; Kwon, J.-D.; Hong, S.-W.; Lee, C.-S.; Ryu, H.-Y.; Hong, S.; Kim, J.; Heo, T.-Y.; Cho, B. Modulation of Synaptic Plasticity Mimicked in Al Nanoparticle-Embedded IGZO Synaptic Transistor. *Advanced Electronic Materials* **2020**, *6* (4), 1901072.
(127) He, Y.; Liu, R.; Jiang, S.; Chen, C.; Zhu, L.; Shi, Y.; Wan, Q. IGZO-based floating-gate synaptic transistors for neuromorphic computing. *J. Phys. D: Appl. Phys.* **2020**, *53* (21), 215106.
(128) Wang, S.; He, C.; Tang, J.; Lu, X.; Shen, C.; Yu, H.; Du, L.; Li, J.; Yang, R.; Shi, D.; Zhang, G. New floating gate memory with excellent retention characteristics. *Advanced Electronic Materials* **2019**, *5* (4), 1800726.
(129) Ji, H.; Wei, Y.; Zhang, X.; Jiang, R. Homogeneous-oxide stack in IGZO thin-film transistors for multi-level-cell NAND memory application. *Appl. Phys. Lett.* **2017**, *111* (20), 202102.
(130) Hwang, E. S.; Kim, J. S.; Jeon, S. M.; Lee, S. J.; Jang, Y.; Cho, D.-Y.; Hwang, C. S. In2Ga2ZnO7 oxide semiconductor based charge trap device for NAND flash memory. *Nanotechnology* **2018**, *29* (15), 155203.

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

(131) Na, S.-Y.; Yoon, S.-M. Impacts of HfO 2/ZnO Stack-Structured Charge-Trap Layers Controlled by Atomic Layer Deposition on Nonvolatile Memory Characteristics of In-Ga-Zn-O Channel Charge-Trap Memory Thin-Film Transistors. *IEEE Journal of the Electron Devices Society* **2019**, *7*, 453−461.
(132) Yun, D.-J.; Kang, H.-B.; Yoon, S.-M. Process optimization and device characterization of nonvolatile charge trap memory transistors using In−Ga−ZnO thin films as both charge trap and active channel layers. *IEEE Trans. Electron Devices* **2016**, *63* (8), 3128−3134.
(133) Yin, H.; Hyun, J.; Park, Y.; Kim, S.; Lim, H.; Min, Y.; Kim, C. J.; Song, I.; Park, J.; Kim, S.-W.; Tikhonovsky, A. Program/erase characteristics of amorphous gallium indium zinc oxide nonvolatile memory. *IEEE Trans. Electron Devices* **2008**, *55* (8), 2071−2077.
(134) Liu, D.-D.; Liu, W.-J.; Pei, J.-X.; Xie, L.-Y.; Huo, J.; Wu, X.; Ding, S.-J. Voltage-Polarity Dependent Programming Behaviors of Amorphous In−Ga−Zn−O Thin-Film Transistor Memory with an Atomic-Layer-Deposited ZnO Charge Trapping Layer. *Nanoscale Res. Lett.* **2019**, *14* (1), 1−8.
(135) Chen, S.; Zhang, W.-P.; Cui, X.-M.; Ding, S.-J.; Sun, Q.-Q.; Zhang, W. Monochromatic light-assisted erasing effects of In-Ga-Zn-O thin film transistor memory with Al2O3/Zn-doped Al2O3/Al2O3 stacks. *Appl. Phys. Lett.* **2014**, *104* (10), 103504.
(136) Liu, D.-D.; Pei, J.; Li, L.; Huo, J.; Wu, X.; Liu, W.-J.; Ding, S.-J. Multilevel memory and synaptic characteristics of a-IGZO thin-film transistor with atomic layer−deposited Al2O3/ZnO/Al2O3 stack layers. *J. Mater. Res.* **2020**, *35* (7), 732−737.
(137) Ryoo, H.-J.; Yoon, S.-M. Channel conductance modulation of dual-gate charge-trap oxide synapse TFT using In-Ga-Zn-O channel and ZnO trap layers. *IEEE Electron Device Lett.* **2020**, *41* (11), 1661−1664.
(138) Liu, D.-D.; Pei, J.; Li, L.; Huo, J.; Wu, X.; Liu, W.-J.; Ding, S.-J. Multilevel memory and synaptic characteristics of a-IGZO thin-film transistor with atomic layer−deposited Al2O3/ZnO/Al2O3 stack layers. *J. Mater. Res.* **2020**, *35* (7), 732−737.
(139) Perera, R.; Ikeda, A.; Hattori, R.; Kuroki, Y. Trap assisted leakage current conduction in thin silicon oxynitride films grown by rapid thermal oxidation combined microwave excited plasma nitridation. *Microelectron. Eng.* **2003**, *65* (4), 357−370.
(140) Bak, J. Y.; Ryu, M.-K.; Park, S. H. K.; Hwang, C.-S.; Yoon, S. M. Impact of charge-trap layer conductivity control on device performances of top-gate memory thin-film transistors using IGZO channel and ZnO charge-trap layer. *IEEE Trans. Electron Devices* **2014**, *61* (7), 2404−2411.
(141) Wang, T.-Y.; Meng, J.-L.; He, Z.-Y.; Chen, L.; Zhu, H.; Sun, Q.-Q.; Ding, S.-J.; Zhou, P.; Zhang, D. W. Ultralow Power Wearable Heterosynapse with Photoelectric Synergistic Modulation. *Advanced Science* **2020**, *7* (8), 1903480.
(142) Yu, X.; Marks, T. J.; Facchetti, A. Metal oxides for optoelectronic applications. *Nat. Mater.* **2016**, *15* (4), 383−396.
(143) Liu, Y.; Weiss, N. O.; Duan, X.; Cheng, H.-C.; Huang, Y.; Duan, X. Van der Waals heterostructures and devices. *Nature Reviews Materials* **2016**, *1* (9), 16042.
(144) Wang, Q. H.; Kalantar-Zadeh, K.; Kis, A.; Coleman, J. N.; Strano, M. S. Electronics and optoelectronics of two-dimensional transition metal dichalcogenides. *Nat. Nanotechnol.* **2012**, *7* (11), 699−712.
(145) Ahmed, T.; Tahir, M.; Low, M. X.; Ren, Y.; Tawfik, S. A.; Mayes, E. L. H.; Kuriakose, S.; Nawaz, S.; Spencer, M. J. S.; Chen, H.; Bhaskaran, M.; Sriram, S.; Walia, S. Fully Light-Controlled Memory and Neuromorphic Computation in Layered Black Phosphorus. *Adv. Mater.* **2021**, *33* (10), 2004207.
(146) Garcia, V.; Bibes, M. Ferroelectric tunnel junctions for information storage and processing. *Nat. Commun.* **2014**, *5* (1), 1−12.
(147) Khan, A. I.; Keshavarzi, A.; Datta, S. The future of ferroelectric field-effect transistor technology. *Nature Electronics* **2020**, *3* (10), 588−597.
(148) Kwon, D.; Cheema, S.; Shanker, N.; Chatterjee, K.; Liao, Y.-H.; Tan, A. J.; Hu, C.; Salahuddin, S. Negative capacitance FET with 1.8-nm-thick Zr-doped HfO2 oxide. *IEEE Electron Device Lett.* **2019**, *40* (6), 993−996.
(149) Lehninger, D.; Ellinger, M.; Ali, T.; Li, S.; Mertens, K.; Lederer, M.; Olivio, R.; Kampfe, T.; Hanisch, N.; Biedermann, K.; Rudolph, M.; Brackmann, V.; Sanctis, S.; Jank, M. P. M.; Seidel, K. A Fully Integrated Ferroelectric Thin-Film-Transistor−Influence of Device Scaling on Threshold Voltage Compensation in Displays. *Advanced Electronic Materials* **2021**, *7*, 2100082.
(150) Kim, E.; Kim, K.; Yoon, S. Investigation of the ferroelectric switching behavior of P (VDF-TrFE)-PMMA blended films for synaptic device applications. *J. Phys. D: Appl. Phys.* **2016**, *49* (7), 075105.
(151) Cho, Y.; Ahn, D.; Park, J. B.; Pak, S.; Lee, S.; Jun, B. O.; Hong, J.; Lee, S. Y.; Jang, J. E.; Hong, J.; Morris, S. M.; Sohn, J. I.; Cha, S. N.; Kim, J. M. Enhanced Ferroelectric Property of P (VDF-TrFE-CTFE) Film Using Room-Temperature Crystallization for High-Performance Ferroelectric Device Applications. *Advanced Electronic Materials* **2016**, *2* (10), 1600225.
(152) Izyumskaya, N.; Alivov, Y.-I.; Cho, S.-J.; Morkoç, H.; Lee, H.; Kang, Y.-S. Processing, structure, properties, and applications of PZT thin films. *Critical reviews in solid state and materials sciences* **2007**, *32* (3−4), 111−202.
(153) Besleaga, C.; Radu, R.; Balescu, L.-M.; Stancu, V.; Costas, A.; Dumitru, V.; Stan, G.; Pintilie, L. Ferroelectric field effect transistors based on PZT and IGZO. *IEEE Journal of the Electron Devices Society* **2019**, *7*, 268−275.
(154) Zhong, G.; Zi, M.; Ren, C.; Xiao, Q.; Tang, M.; Wei, L.; An, F.; Xie, S.; Wang, J.; Zhong, X.; Huang, M.; Li, J. Flexible electronic synapse enabled by ferroelectric field effect transistor for robust neuromorphic computing. *Appl. Phys. Lett.* **2020**, *117* (9), 092903.
(155) Pintilie, L.; Vrejoiu, I.; Hesse, D.; Alexe, M. Polarization fatigue and frequency-dependent recovery in Pb (Zr, Ti) O 3 epitaxial thin films with $SrRuO_3$ electrodes. *Applied physics letters* **2006**, *88* (10), 102908.
(156) Park, M. H.; Lee, Y. H.; Mikolajick, T.; Schroeder, U.; Hwang, C. S. Review and perspective on ferroelectric HfO2-based thin films for memory applications. *MRS Commun.* **2018**, *8* (3), 795−808.
(157) Gaskins, J. T.; Hopkins, P. E.; Merrill, D. R.; Bauers, S. R.; Hadland, E.; Johnson, D. C.; Koh, D.; Yum, J. H.; Banerjee, S.; Nordell, B. J.; Paquette, M. M.; Caruso, A. N.; Lanford, W. A.; Henry, P.; Ross, L.; Li, H.; Li, L.; French, M.; Rudolph, A. M.; King, S. W. Investigation and review of the thermal, mechanical, electrical, optical, and structural properties of atomic layer deposited high-k dielectrics: beryllium oxide, aluminum oxide, hafnium oxide, and aluminum nitride. *ECS Journal of Solid State Science and Technology* **2017**, *6* (10), N189.
(158) Böscke, T.; Müller, J.; Bräuhaus, D.; Schröder, U.; Böttger, U. Ferroelectricity in hafnium oxide thin films. *Appl. Phys. Lett.* **2011**, *99* (10), 102903.
(159) Polakowski, P.; Müller, J. Ferroelectricity in undoped hafnium oxide. *Appl. Phys. Lett.* **2015**, *106* (23), 232905.
(160) Cheema, S. S.; Kwon, D.; Shanker, N.; Dos Reis, R.; Hsu, S.-L.; Xiao, J.; Zhang, H.; Wagner, R.; Datar, A.; McCarter, M. R. Enhanced ferroelectricity in ultrathin films grown directly on silicon. *Nature* **2020**, *580* (7804), 478−482.
(161) Park, M. H.; Lee, Y. H.; Mikolajick, T.; Schroeder, U.; Hwang, C. S. Thermodynamic and kinetic origins of ferroelectricity in fluorite structure oxides. *Advanced Electronic Materials* **2019**, *5* (3), 1800522.
(162) Kim, M.-K.; Lee, J.-S. Ferroelectric analog synaptic transistors. *Nano Lett.* **2019**, *19* (3), 2044−2050.
(163) Kim, M. K.; Lee, J. S. Synergistic Improvement of Long-Term Plasticity in Photonic Synapses Using Ferroelectric Polarization in Hafnia-Based Oxide-Semiconductor Transistors. *Adv. Mater.* **2020**, *32* (12), 1907826.
(164) Jerry, M.; Chen, P. Y.; Zhang, J.; Sharma, P.; Ni, K.; Yu, S.; Datta, S. Ferroelectric FET analog synapse for acceleration of deep neural network training. *In 2017 IEEE International Electron Devices Meeting (IEDM)* **2017**, 6.2.1−6.2.4.

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448

(165) Saha, A.; Islam, A. N. M. N.; Zhao, Z.; Deng, S.; Ni, K.; Sengupta, A. Intrinsic synaptic plasticity of ferroelectric field effect transistors for online learning. *Appl. Phys. Lett.* **2021**, *119* (13), 133701.
(166) Seo, M.; Kang, M. H.; Jeon, S. B.; Bae, H.; Hur, J.; Jang, B. C.; Yun, S.; Cho, S.; Kim, W. K.; Kim, M. S.; Hwang, K. M.; Hong, S.; Choi, S. Y.; Choi, Y. K. First Demonstration of a Logic-Process Compatible Junctionless Ferroelectric FinFET Synapse for Neuromorphic Applications. *IEEE Electron Device Lett.* **2018**, *39* (9), 1445−1448.
(167) Xi, F.; Han, Y.; Liu, M.; Bae, J. H.; Tiedemann, A.; Grützmacher, D.; Zhao, Q.-T. Artificial Synapses Based on Ferroelectric Schottky Barrier Field-Effect Transistors for Neuromorphic Applications. *ACS Appl. Mater. Interfaces* **2021**, *13* (27), 32005−32012.
(168) Mo, F.; Saraya, T.; Hiramoto, T.; Kobayashi, M. Reliability characteristics of metal/ferroelectric-HfO2/IGZO/metal capacitor for non-volatile memory application. *Applied Physics Express* **2020**, *13* (7), 074005.
(169) Mo, F.; Tagawa, Y.; Jin, C.; Ahn, M.; Saraya, T.; Hiramoto, T.; Kobayashi, M. Low-voltage operating ferroelectric FET with ultrathin IGZO channel for high-density memory application. *IEEE Journal of the Electron Devices Society* **2020**, *8*, 717−723.
(170) Mo, F.; Tagawa, Y.; Jin, C.; Ahn, M.; Saraya, T.; Hiramoto, T.; Kobayashi, M. Experimental Demonstration of Ferroelectric HfO2 FET with Ultrathin-body IGZO for High-Density and Low-Power Memory Application. In *2019 Symposium on VLSI Technology*; IEEE, 2019; pp T42−T43.
(171) Cao, R.; Wang, Y.; Zhao, S.; Yang, Y.; Zhao, X.; Wang, W.; Zhang, X.; Lv, H.; Liu, Q.; Liu, M. Effects of Capping Electrode on Ferroelectric Properties of Hf0.5Zr0.5O_2 Thin Films. *IEEE Electron Device Lett.* **2018**, *39* (8), 1207−1210.
(172) Ryu, H.; Xu, K.; Kim, J.; Kang, S.; Guo, J.; Zhu, W. Exploring new metal electrodes for ferroelectric aluminum-doped hafnium oxide. *IEEE Trans. Electron Devices* **2019**, *66* (5), 2359−2364.
(173) Goh, Y.; Cho, S. H.; Park, S. K.; Jeon, S. Oxygen vacancy control as a strategy to achieve highly reliable hafnia ferroelectrics using oxide electrode. *Nanoscale* **2020**, *12* (16), 9024−9031.
(174) Lu, T.; Liang, R.; Zhao, R.; Yang, Y.; Ren, T.-L. Fabrication and Characterization of Ferroelectric HfZrO-based Synaptic Transistors with Multi-state Plasticity. In *2020 4th IEEE Electron Devices Technology & Manufacturing Conference (EDTM)*; IEEE, 2020; pp 1−4.
(175) Zhao, R.; Zhao, X.; Liu, H.; Shao, M.; Feng, Q.; Liu, T.; Lu, T.; Wu, X.; Yi, Y.; Ren, T.-L. Reconfigurable logic-memory hybrid device based on ferroelectric $Hf_{0.5}Zr_{0.5}O_2$. *IEEE Electron Device Lett.* **2021**, *42* (8), 1164−1167.
(176) Duan, H.; Liang, L.; Wu, Z.; Zhang, H.; Huang, L.; Cao, H. IGZO/CsPbBr3-Nanoparticles/IGZO Neuromorphic Phototransistors and Their Optoelectronic Coupling Applications. *ACS Appl. Mater. Interfaces* **2021**, *13* (25), 30165−30173.
(177) Kwon, S. M.; Cho, S. W.; Kim, M.; Heo, J. S.; Kim, Y. H.; Park, S. K. Environment-Adaptable Artificial Visual Perception Behaviors Using a Light-Adjustable Optoelectronic Neuromorphic Device Array. *Adv. Mater.* **2019**, *31* (52), 1906433.
(178) Hong, S.; Cho, H.; Kang, B. H.; Park, K.; Akinwande, D.; Kim, H. J.; Kim, S. Neuromorphic Active Pixel Image Sensor Array for Visual Memory. *ACS Nano* **2021**, *15* (9), 15362−15370.
(179) Yu, J.; Liang, L.; Hu, L.; Duan, H.; Wu, W.; Zhang, H.; Gao, J.; Zhuge, F.; Chang, T.; Cao, H. Optoelectronic neuromorphic thin-film transistors capable of selective attention and with ultra-low power dissipation. *Nano Energy* **2019**, *62*, 772−780.
(180) Yang, J.; Kwak, H.; Lee, Y.; Kang, Y.-S.; Cho, M.-H.; Cho, J. H.; Kim, Y.-H.; Jeong, S.-J.; Park, S.; Lee, H.-J.; Kim, H. MoS2−InGaZnO Heterojunction Phototransistors with Broad Spectral Responsivity. *ACS Appl. Mater. Interfaces* **2016**, *8* (13), 8576−8582.
(181) Wu, Q.; Wang, J.; Cao, J.; Lu, C.; Yang, G.; Shi, X.; Chuai, X.; Gong, Y.; Su, Y.; Zhao, Y.; Lu, N.; Geng, D.; Wang, H.; Li, L.; Liu, M. Photoelectric plasticity in oxide thin film transistors with tunable synaptic functions. *Advanced Electronic Materials* **2018**, *4* (12), 1800556.
(182) Berweger, S.; Zhang, H.; Sahoo, P. K.; Kupp, B. M.; Blackburn, J. L.; Miller, E. M.; Wallis, T. M.; Voronine, D. V.; Kabos, P.; Nanayakkara, S. U. Spatially Resolved Persistent Photoconductivity in MoS2−WS2 Lateral Heterostructures. *ACS Nano* **2020**, *14* (10), 14080−14090.
(183) Choi, K.-H.; Koo, H.-W.; Kim, T.-W.; Kim, H.-K. Antireflective ZnSnO/Ag bilayer-based transparent source and drain electrodes for transparent thin film transistors. *Appl. Phys. Lett.* **2012**, *100* (26), 263505.

https://doi.org/10.1021/acsaelm.1c01088
ACS Appl. Electron. Mater. 2022, 4, 1427−1448